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Adaptive update-based digital twin model and dual-domain collaborative lightweight adversarial transfer network for rolling bearing fault diagnosis

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Abstract

Due to the difference in the distribution of virtual and physical data caused by external influences such as load and rotational speed fluctuations, it is difficult for digital twins to perform data augmentation and knowledge transfer, which results in low diagnostic accuracy for rolling bearings. To alleviate this problem, this study introduces an intelligent diagnostic framework that combines an adaptively updated digital twin (DT) with high-fidelity models and a dual-domain collaborative lightweight adversarial transfer network (DCLATN). This framework facilitates information transfer from physical entities to dynamic models, thereby enhancing cross-domain adaptability for fault diagnosis under complex operating conditions. Firstly, a high-fidelity DT model reflecting the physical operating state of the rolling bearing was established through dynamic modeling and structural parameter updating, and the model was then simulated and analyzed. Then, the DCLATN is constructed to minimize the discrepancy between the source and target domains, where the depth-separable convolution layers with weight-sharing mechanism are constructed for feature extraction and the gradient inversion layers are utilized for feature discrimination. The superiority of the proposed method lies in that it effectively eliminates the domain distribution disparities in vibration signals under diverse operating conditions, thereby ensuring the robustness and efficiency of the model in cross-domain fault diagnosis. The experimental results verify that the proposed method surpasses the state-of-the-art bearing fault diagnosis methods.

Keywords: Rolling bearing; Digital twin; Transfer learning; Intelligent fault diagnosis

1. Introduction

As vital components within sophisticated machinery, rolling-element bearings are frequently subjected to sustained heavy loading and speed fluctuations. Such operational stresses can undermine equipment safety, leading to extended downtime, elevated maintenance expenditures, and, in severe cases, an increased risk of catastrophic failure. Therefore, continuous bearing health monitoring coupled with prompt fault detection is crucial for ensuring both the safety and dependability of machinery operations.

Over the past few years, data-driven techniques, particularly those that center on fault diagnosis through artificial intelligence, such as artificial neural networks (ANNs) [1], support vector machines (SVMs) [2], convolutional neural networks (CNNs) [3], and deep neural networks (DNNs) [4], have garnered significant interest both in academic circles and within the industrial

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sector. The convolutional neural network serves as a quintessential example of deep learning [5]. It automatically extracts deep features from raw data through the use of convolutional layers, pooling layers, and fully connected layers. With the help of adversarial training techniques, the model's ability to adapt to distributional differences between domains, which addresses challenges in cross-domain learning, including distributional differences and label scarcity in the target domain. In complex and variable operating environments with data imbalance, adversarial neural networks can learn the potential distribution patterns of the data, allowing the model to perform fault diagnosis stably under different conditions and effectively avoid overfitting.[6-9]. However, most existing approaches tackle only one of these issues either aligning distributions across domains or compensating for scarce labels rather than providing a unified framework that simultaneously addresses both domain shifts and target domain annotation gaps.

In light of these challenges, researchers have started exploring approaches to tackle the complexities of bearing fault diagnosis across different domains and the scarcity of labeled data in the source domain. By combining digital twin methodologies with uncertainty quantification modeling, cross-domain fault diagnosis can effectively overcome both label scarcity and domain mismatch issues in mechanical diagnostics. Zhou et al. [10] propose an entropy-weighted stream alignment that uses label entropy to dynamically reweight samples, thereby reducing distribution variance, mitigating imbalances, and improving diagnostic accuracy and robustness. She et al. [11] A proposed a fault diagnosis method for combine harvester gearboxes based on meta-transfer learning, which also introduced a multi-step loss optimization method to address the problem of unstable update gradients during training. Qin et al. [12] introduced a multi-source domain adaptation network, which effectively extended the fault samples for model training through the generation of an adversarial network variant. By combining the integrated model of transformer, researchers can extract features from different source domains, which enhances cross-domain feature generalization. Fatemeh Negar Irani et al. [13] introduced a transfer learning framework for deep neural networks that reliably predicts system behaviour under various scenarios and designs structured fault residuals. Wang et al. [14] proposed a HVAC FDD transfer learning approach that facilitates the training of the FDD model by simply plugging in the small adapter module as a transfer learning strategy. Ding et al. [15] presented a deep imbalance domain adaptation framework to tackle the labeling bias of category imbalance and strengthen the cross-domain generalization ability. Lu et al. [16] proposed a wavelet packet energy-based embedded autoencoder (WPEEAE) combined with a dynamic weighting strategy to address the issue of inaccessible target domain data in fault diagnosis. Lu et al. [17] proposed the adaptive single-source open domain generalisation network (ASODGN) to address the challenge of handling data from multiple source domains in practical engineering applications, where the label types of source and target domain data are consistent. Lu et al. [18] proposed a dynamic hybrid distribution alignment mechanism within the deep dynamic hybrid domain adaptation network (DDHDAN). This enhances diagnostic capability by minimising hybrid domain distribution

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discrepancies between fault-frequency-related physical features.

However, the fundamental issue of insufficient fault information or related measured data remains unresolved for actual machinery equipment. By combining digital twin methodologies with uncertainty quantification modeling, cross-domain fault diagnosis can effectively overcome both label scarcity and domain mismatch issues in mechanical diagnostics because genuine fault examples are exceedingly rare in real equipment (machines normally run without failure for long periods), fault injection tests are costly and pose safety risks, and comprehensive sensor deployment is often impractical, all of which severely limit the availability of measured fault data.

Recently, digital twin technology is gaining prominence in modern engineering and industrial applications, which builds a virtual replica of physical equipment to generate simulation signals and expand fault samples. To conform to real-world operational conditions, physics-based constraints and cross-domain alignment techniques are applied in digital twin technology. Habbouche et al. [19] developed high-fidelity digital twins that obviate reliance on destructive experimentation for generating fault scenarios and concurrently alleviate inherent operational constraints in industrial applications. Ming et al. [20] presented a digital twin-assisted framework that facilitates the transfer of information from a physical entity to a dynamic model,which generates digital twins to enhance the data. Huang et al [21] constructed a bearing digital twin model to generate enough low-fidelity analog data, analog data so as to improve feature enhancement using limited high-fidelity analog data and hence class incremental learning. Kohtz et al. [22] developed FEA model for simulation of faults as digital twins,used distance clustering for identifying various fault modes. Yan et al. [23] introduced a DT assisted framework for unbalanced fault diagnosis. Xu et al. [24] devoloped a graph transfer learning based digital twin model. Nonetheless, prevailing digital twin methodologies are designed for systems operating under steady state or predictably varying regimes and thus prove inadequate for a non-stationary operating condition. In response, several enhanced digital twin approaches have been proposed to better accommodate non-stationary operating conditions. Wen et al. [25] takes into account improved fatigue theory with nonlinear load interactions, real-time stress measurements with multiple correction factors by integrating a fatigue life predictor driven by a nonlinear digital twin. Angelis M et al. [26] proposes an online update method for digital twins based on non-parametric resonant random sets. Booyse W et al. [27] proposed the deep digital twin framework, which enables real time asset health assessment through a deep generative model that learns health state distributions from only healthy operational data at the beginning of an asset's lifecycle, eliminating the need for historical failure data and manual feature engineering.

Although digital twin technology shows great potential, it still faces many challenges when dealing with cross-domain fault diagnosis tasks with missing labels. Traditional digital twin models do not take into account the friction effect in bearing operation, and the resulting nonlinear vibration can overlap with the shock spectrum of early fault features, leading to noise swamping of fault features in the time-frequency analysis, which in turn triggers data distortion. Friction

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induces cyclic contact force variations between rolling elements and raceways, causing abnormal bearing vibrations. Shock impulses generated as rolling bodies traverse defects then create mismatches between simulated and real world signals.

To address these problems, a spring-damper system is introduced into the digital twin model to alleviate the data distortion caused by abnormal signal fluctuations, shock pulses and friction effects. We first build a spring-damper vibration coupling model of a rolling bearing to generate signals under different health states. Then, we introduce a dual-domain collaborative lightweight adversarial transfer network (DCLATN) for robust cross-domain bearing fault diagnosis, which employs depthwise separable convolutions and a gradient reversal layer with a domain discriminator to realize domain invariant representations. A framework based on digital twin model and dual-domain collaborative lightweight adversarial transfer networkis proposed to integrate a digital twin model with a dual domain collaborative lightweight adversarial transfer network, which tackles performance degradation of cross-domain bearing fault diagnosis with large distribution gaps and scarce labeled data.

The main contributions of this study include

(1) This study introduces an advanced digital twin architecture to achieve cross-domain adaptation alignment, which integrates physical modeling with data-driven methods. It enables the migration of simulation insights into real-world settings subject to time-varying dynamics and alleviates the paucity of annotated real-world datasets for deep neural network training, thereby markedly enhancing model robustness and accuracy across a range of operational conditions.

(2) An advanced high-fidelity digital twin-based multi-physics vibration model for rolling bearings has been formulated to generate synthetic datasets encompassing diverse fault modes under nonlinear regimes. The model accommodates bearings of varying designs, geometries, and operational parameters, rendering it applicable across a broad spectrum of industrial contexts.

(3) A DCLATN integrating a gradient inversion layer and a discriminator is proposed to reduce distribution discrepancies of bearing data under varying conditions and enhance the fault feature invariance in the source and target domains, which provides efficient and accurate cross-domain feature extraction.

(4) The effectiveness and robustness of the proposed method are verified through experiments on two cases, which show that the accuracy of the method is significantly higher compared with traditional fault diagnosis methods.

The rest of the paper is organised as follows. Section 2 introduces the digital twin model for fault diagnosis and data construction. Section 3 describes how to construct a dual-domain collaborative lightweight adversarial transfer network framework. Section 4 verifies the data reliability of the twin model. Section 5 demonstrates the method's effectiveness through experiments and compares it with several other approaches. Finally, Section 6 concludes this study.

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2. Digital twin

2.1 Bearing simulation model

This data imbalance makes it difficult to accurately identify fault conditions even with highly accurate diagnostic models. To solve the above challenges, we constructed a rolling bearing digital twin model. The model is capable of simulating various fault conditions in a virtual environment, and thus generates large-scale data sets covering multiple health states. The generated data can be used to train the diagnostic model, which not only effectively alleviates the data imbalance problem, but also reduces the cost and practical difficulty of collecting fault data in the real environment.

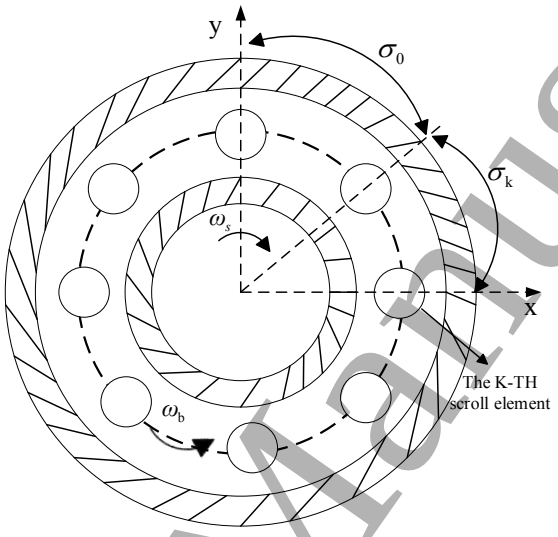


Fig. 1. Structure of the physical running model of the twin model

During the operation of the DT model, it is assumed that the internal raceway rotates with the spindle, while the external raceway remains fixed to the bearing housing. The rolling bodies moves rotationally between the internal and external raceways [28,29]. The rotational frequency is denoted as f_s, and its angular velocity. This may be computed by applying the subsequent formula

The rotational angular velocity of the cage between the internal and external raceways is

$$\omega_i = \pi * f_s (1 - \frac{d_b}{D_m} \cos\alpha) \tag{1}$$

where, d_b represents the rolling bodies diameter, D_m represents the bearing internal ring pitch circle diameter.

The azimuth angle of the k-th rolling bodies in the radial plane of the bearing can be expressed as

$$\sigma_k = \omega_i t + \frac{2\pi(k-1)}{N} + \sigma_0 \tag{2}$$

where σ_0 is the initial azimuth angle of the first rolling bodies, and N is the number of rolling bodies in the bearing.

2.2 Rolling body operation

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This work primarily targets high-speed rolling bearings. therefore, high-speed effects must be considered. For the *k-th* rolling bodies, the force analysis is shown in the Fig. 2 below

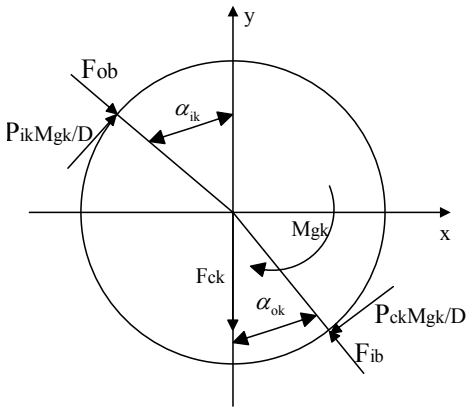


Fig. 2. Equilibrium of forces on a rolling body

This diagram enables the computation of the rotational forces and moments acting on the rolling elements. The dynamic equations of the rolling bodies can be derived as

$$F_{ib}\sin\alpha_{ik} - F_{ob}\sin\alpha_{ok} = \frac{P_{ok}M_{gk}\cos\alpha_{ok}}{D} - \frac{P_{ik}M_{gk}\cos\alpha_{ik}}{D} \tag{3}$$

$$F_{ib}\cos\alpha_{ik} - F_{ob}\cos\alpha_{ok} + F_{ck} = \frac{P_{ik}M_{gk}\sin\alpha_{ik}}{D} - \frac{P_{ok}M_{gk}\sin\alpha_{ok}}{D} \tag{4}$$

where F_{ck} is the centrifugal force on the *k-th* online element, M_{gk} is the gyroscopic moment, F_{ib} is the contact force between the bearing internal raceway and the rolling bodies, and F_{ob} represents the contact force between the bearing external ring and the rolling bodies.

m_b is the mass of the *k-th* rolling body, which can be calculated as

$$m_b = \frac{\rho\pi d_d^3}{6} \tag{5}$$

where ρ is the rolling body material density, the *k-th* rolling body centrifugal force F_{ck} can be calculated as

$$F_{ck} = m_b D_m \omega_b^2 \tag{6}$$

The *k-th* rolling body momentum B can be calculated as

$$B = \frac{\rho\pi d_b}{60} \tag{7}$$

In addition, the *k-th* rolling body gyroscopic moment M_{gk} can be calculated as

$$M_{gk} = B\omega_s\omega_b\sin\alpha_k \tag{8}$$

When a rolling body passes through a local defect in a bearing component, a collision occurs, generating a narrow, short-duration pulse excitation. This extremely brief pulse excitation distributes energy across a wide frequency band, thereby exciting the natural frequencies of various bearing components and causing vibration.

Friction is a key parameter affecting operational lifespan and also serves as an indicator of potential faults. Friction causes the actual motion state of the components inside the bearing to

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differ from the theoretical value under the frictionless assumption, which affects the simulation accuracy of the vibration signals[30]. The derivation of the calculation is as

$$F_f = c \cdot v + F_n \cdot I \tag{9}$$

where F_f represents the friction force, I represents the velocity-dependent friction coefficient, F_n represents the normal contact force, c is the coefficient of viscous damping, and v represents the relative velocity of the contact surface.

Rolling bearings are precision mechanical components designed to reduce friction loss between shafts and bearing housings. When a bearing component fails, it repeatedly impacts other components during subsequent operation cycles, generating instantaneous and periodic vibration impacts. The fault frequency can be calculated based on the physical parameters of the rolling bearing. Different fault severities result in varying fault frequencies [31]. Based on the physical structure of rolling bearings, the fault characteristic frequencies for various bearing components can be determined using the following equations.

The internal raceway fault frequency of rolling bearings is

$$f_i = \frac{N\omega_i}{4\pi}(1 + \frac{d}{D_i}\cos\alpha) \tag{10}$$

The rolling bodies fault frequency of rolling bearings is

$$f_e = \frac{D_i\omega_i}{4\pi d}\{1 + (\frac{d}{D_i}\cos\alpha)^2\} \tag{11}$$

The external raceway fault frequency of rolling bearings is

$$f_o = \frac{N\omega_i}{4\pi}(1 - \frac{d}{D_i}\cos\alpha) \tag{12}$$

Here, N represents the number of bearing rolling bodies, d denotes the internal diameter of the bearing, D signifies the external diameter of the bearing, α is the contact angle between the rolling bodies and the raceways, and ω is the angular velocity of the cage.

2.3 Comprehensive model

According to Hertzian contact theory [32], under radial loading, the contact zones between the rolling bodies and the internal/external raceways can be approximated as linear spring-damper elements.

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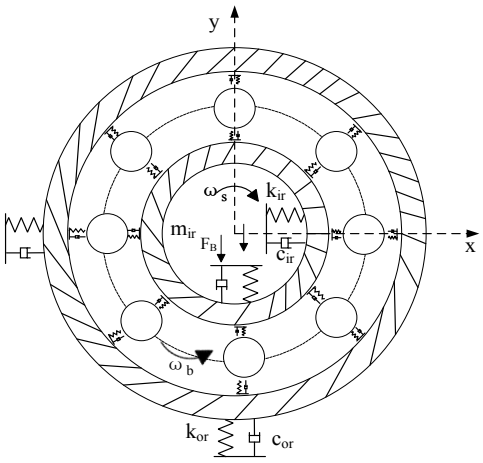


Fig. 3. High fidelity digital twin-based multi-physics vibration model for rolling bearings

In the virtual simulation environment, a high-fidelity digital twin-based multi-physics vibration model for rolling bearings is employed to generate simulated vibration signals. Typically, based on the bearing's specific operational conditions, the internal raceway is assembled with the shaft and the external raceway with the bearing housing using a mechanically tight interference fit, which is designed to ensure structural stability and load transmission efficiency.

Using mechanical vibration theory and the high-fidelity digital twin-based multi-physics vibration model for rolling bearings shown in the Fig. 3, the motion and contact processes between the bearing shaft and internal raceway, rolling bodies and internal/external raceways, and external race and housing are analyzed. Considering the external load F_B, the dynamic differential equation group for rolling bearings is derived as

$$m_{ir}\ddot{x}_{ir} + \sum_{k=1}^{N} c_{ir}(\dot{x}_{ir} - \dot{x}_{bk}) + k_{ir}x_{ir} - F_{BX} = -F_{bx}^{ib} \tag{13}$$

$$m_{or}\ddot{y}_{or} + \sum_{k=1}^{N} c_{or}(\dot{y}_{bk} - \dot{y}_{or}) + k_{or}x_{ir} + F_{by}^{or} = m_{or}\mathrm{g} \tag{14}$$

$$m_{or}\ddot{x}_{or} + \sum_{k=1}^{N} c_{or}(\dot{x}_{bk} - \dot{x}_{or}) + k_{or}y_{or} = F_{bx}^{or} \tag{15}$$

$$m_{ir}\ddot{y}_{ir} + \sum_{k=1}^{N} c_{ir}(\dot{y}_{ir} - \dot{y}_{bk}) + k_{ir}x_{ir} + F_{by}^{ir} = F_{BY} + m_{ir}\mathrm{g} \tag{16}$$

$$\begin{aligned} &m_{\mathrm{b}}\ddot{x}_{\mathrm{bk}} + c_{\mathrm{b}}(\dot{x}_{\mathrm{bk}} - \dot{x}_{or}) + H\left(\varphi_k^{or}\right)k_b\left(\varphi_k^{or}\right)^{3/2}\cos\alpha_{ik}\cos\eta_k \\ &= H\left(\varphi_k^{ir}\right)k_b\left(\varphi_k^{ir}\right)^{3/2}\cos\alpha_{ik}\cos\eta_k + c_{\mathrm{b}}(\dot{x}_{\mathrm{ir}} - \dot{x}_{\mathrm{bk}}) \end{aligned} \tag{17}$$

$$\begin{aligned} &m_{\mathrm{b}}\ddot{x}_{\mathrm{bk}} + c_{\mathrm{b}}(\dot{y}_{\mathrm{bk}} - \dot{y}_{or}) + H\left(\varphi_k^{or}\right)k_b\left(\varphi_k^{or}\right)^{3/2}\cos\alpha_{ik}\cos\eta_k \\ &= H\left(\varphi_k^{ir}\right)k_b\left(\varphi_k^{ir}\right)^{3/2}\cos\alpha_{ik}\sin\eta_k + c_{\mathrm{b}}(\dot{y}_{\mathrm{ir}} - \dot{y}_{\mathrm{bk}}) \end{aligned} \tag{18}$$

The horizontal direction and vertical direction are denoted by x and y, respectively. m_{ir} and m_{or} denote the lumped masses pertaining to the internal component bearing journal and the frame assembly, respectively. c_{ir} and c_{or} are the equivalent contact damping coefficients between the

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internal raceway and shaft and between the external raceway and housing. k_{ir} and k_{or} are the effective contact spring coefficients between the internal raceway and shaft and between the external raceway and housing. c_b and k_b are the comparable contact damping and spring coefficients between the rolling bodies and the internal/external raceways. Variables x_{ir} and y_{ir} denote the inter-raceway displacement components along the x-axes and y-axes, respectively. $\dot{x}_{ir}$ and $\dot{y}_{ir}$ denote the horizontal and vertical velocity components of the inner circle, respectively. The inner circle's acceleration is decomposed into horizontal and vertical projections, denoted by $\ddot{x}_{ir}$ and $\ddot{y}_{ir}$, respectively. The external ring's displacement along the horizontal axis is represented by x_{or}, whereas y_{or} corresponds to its velocity along the vertical axis. The external ring's displacement along the horizontal axis is represented by $\dot{x}_{or}$, whereas $\dot{y}_{or}$ corresponds to its displacement along the vertical axis. The external ring's acceleration along the horizontal axis is represented by $\ddot{x}_{or}$, whereas $\ddot{y}_{or}$ corresponds to its displacement along the vertical axis. The rolling bodies displacement along the horizontal axis is represented by x_b, whereas y_b corresponds to its displacement along the vertical axis. The rolling bodies velocity along the horizontal axis is represented by $\dot{x}_b$, whereas $\dot{y}_{\mathrm{b}}$ corresponds to its displacement along the vertical axis. Lastly, φ_{ik} and φ_{ok} denote the contact deformations between the *k-th* rolling bodies and the internal raceway and external raceway, respectively.

The fourth-order Runge-Kutta method is employed to address the aforementioned equations, yielding the vibration responses of the rolling bodies, internal raceway, and external raceway. The software platform is based on MATLAB R2022b, with the ode45 solver employed to compute the dynamic responses of faulty bearings.

2.4 cross-domain adaptation alignment

To compensate for the smearing of cyclostationary signatures under non-stationary rotational speeds, we propose an order-frequency dimension cyclic spectral correlation (DCSC) framework that enforces alignment in the physical shaft angle domain. While conventional spectral correlation can expose hidden periodicities in vibration signals, these features become diffuse when speed varies, since their manifestation is intrinsically tied to shaft angle. Our DCSC approach overcomes this by applying two successive fourier transforms to the autocorrelation of an angle-synchronized signal. First, a short-time fourier transform (STFT) decomposes the vibration waveform into a time-frequency representation of the carrier spectrum. This time frequency plane is then resampled with respect to the instantaneous shaft angle, thus correcting for speed-induced spectral shifts. Finally, the autocorrelation of this angle-resampled data undergoes an angular fourier transform, producing an order frequency DCSC map whose axes are cyclic modulation order and carrier frequency. By cascading temporal and angular fourier analyses, the method effectively decouples speed variations from the cyclostationary content, yielding a sharp,

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bi-dimensional spectral portrait of fault-related periodic components. To accomplish this, the vibration signal is initially transformed to match the accumulated angle as time progresses.

$$S_o(\theta) = S(x(t)) \tag{19}$$

where $\theta(t)=\int_0^1 \vartheta(t)dt$ represents the accumulated angle, with $\vartheta(t)$ represents the instantaneous angular velocity.

The instantaneous autocorrelation function for a second-order cyclostationary signal is defined as follows

$$T_x(\theta,\varepsilon) = F[S_o(\theta)\cdot S_o{}^*(\theta-\varepsilon)] \tag{20}$$

where $F[\cdot]$ denotes the ensemble average operator, and $*$ signifies the complex conjugate.

Finally, the dual fourier Transform (FT) of $T_x(\theta,\varepsilon)$ transitions the signal into the order-frequency spectral correlation domain, defined as

$$R_x(\alpha,f) = \lim_{k\to\infty}\frac{1}{K}F\{\chi_{\theta\to\alpha,\varepsilon\to f}[S_o(\theta)\cdot S_o{}^*(\theta-\varepsilon)]\} \tag{21}$$

where α and f denote order and frequency, respectively. This generates an order-frequency DCSC map, derived from the cyclic modulation spectrum estimator. The two-dimensional DCSC representation offers a robust basis for aligning data across operating conditions by retaining invariant, fault-specific cyclic patterns even when speed varies. Consequently, the physics-driven alignment stage can extract and emphasize these enduring periodic signatures otherwise masked by rotational fluctuations thereby improving the mode's capacity to capture and transfer underlying fault-related dynamics between different domains.

2.5Particle swarm optimization (PSO)

Swarm intelligence algorithms, modeled after the collaborative behaviors observed in animals, have demonstrated their effectiveness in solving optimization problems. Particle Swarm Optimization (PSO), each particle represents a candidate solution in, where p represents the quantity of unknown variables that need to be optimized. The position of the i-th particle is denoted as the vector t_i. The entire swarm of candidate solutions can be represented as $\mathrm{t}=\{\mathrm{t}_1,\mathrm{t}_2,\mathrm{t}_3,...,\mathrm{t}_Z\}$. To find the optimal solution, the particles' positions are updated iteratively, based on both their individual best performance and the best performance of their neighbors, as described in Equation (23). This iterative update process, driven by both individual experience and collective cooperation, gradually leads the algorithm towards the global optimum.

$$x_i(k+1) = x_i(k) + \mathrm{v}_i(k+1) \tag{22}$$

where k and $k+1$ are the indices of two consecutive iterations and v_i is the velocity vector of the ith particle. These velocity vectors consist of the following three components. (1) The inertia or momentum term helps prevent the particle from making abrupt direction changes; (2) the cognitive term reflects the particle's tendency to move towards its own previous best position; and (3) the social term indicates the particle's inclination to approach the best position found by the

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entire swarm. Considering these three factors, the velocity of the i-th particle is expressed in Equation (24).

$$v_i(k+1) = \omega v_i(k) + m_1(p_i - x_i(k))R_1 + m_2(g - x_i(k))R_2 \quad (23)$$

Where ω is the inertia weight, p_i is the "personal best" of the ith particle; and g represents the "global best", which is the best solution discovered by the entire swarm. The constants m_1 and m_2 referred to as the cognitive and social coefficients, respectively, while $m_1 + m_2 \leq 4$. R_1 and R_2 are two random diagonal matrices, with their elements drawn from a uniform distribution between 0 and 1.

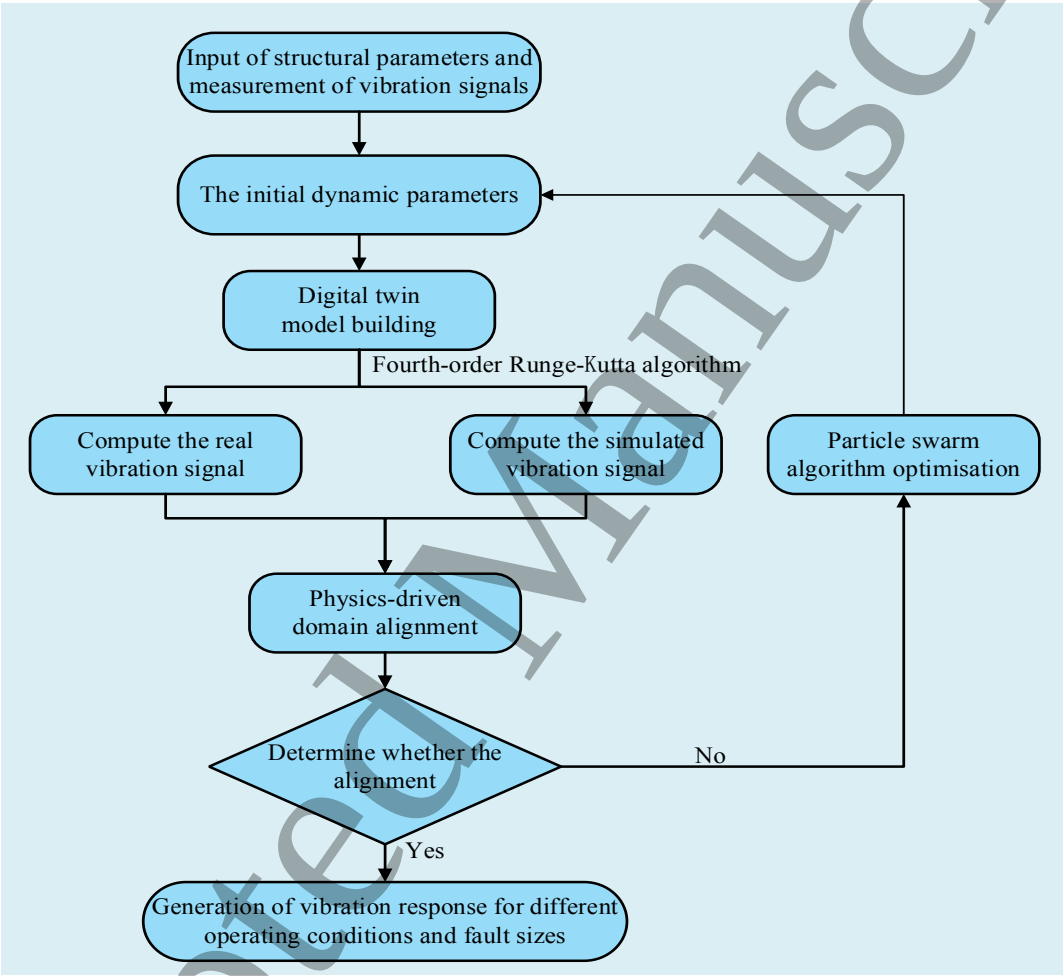


Fig. 4. Flowchart of vibration response generation based on digital twin and physics-driven domain alignment

To delve deeper into the similarity between simulated and measured signals, we compared their effective values and skewness metrics, with the detailed results summarized in Table 2. An analysis of these key statistical characteristics reveals that the deviation between the fault-implanted simulated signals derived from the digital twin model and the real-world fault signals captured by physical sensors is negligible. Thus, these simulated fault signals are sufficiently reliable to serve as a valid data source for training and fine-tuning fault diagnosis models.

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Subsequently, we adopted the bearing fault dataset generated by the pre-established digital twin as the source domain, while using the actually collected bearing status data as the target domain. On this basis, a dual-domain collaborative lightweight adversarial transfer learning network was utilized to implement fault diagnosis on the real-world collected fault signals.

3.Cross-domain fault diagnosis model

3.1 Fault diagnosis via imbalanced transfer learning

In cross condition and multi-device fault diagnosis, transfer learning is widely adopted. The capacity for feature extraction under new operational settings is enhanced by reducing distributional gaps across domains and leveraging empirical evidence acquired within the original domain [33]. This method achieves transferability ability of features by calculating inter-domain statistical moments and incorporating them as complexity regularization terms. The core concepts of this approach are elaborated as follows

$$L_k(M,N) = \| S_M[\Lambda(X^s)] - S_N[\Lambda(X^t)] \|_k^2 \tag{24}$$

where X^s and X^t represents samples from the source and target domains respectively, following the relevant probability distributions M and N. $S[\cdot]$ denotes the mathematical expectation operation. k is the feature mapping space, which is usually used as regenerative kernel Hilbert space (RKHS). $\Lambda(\cdot)$ represents the process of mapping data features. However, most fault diagnosis methods based on transfer learning operate under the idealized assumption of implicitly balanced data classes.

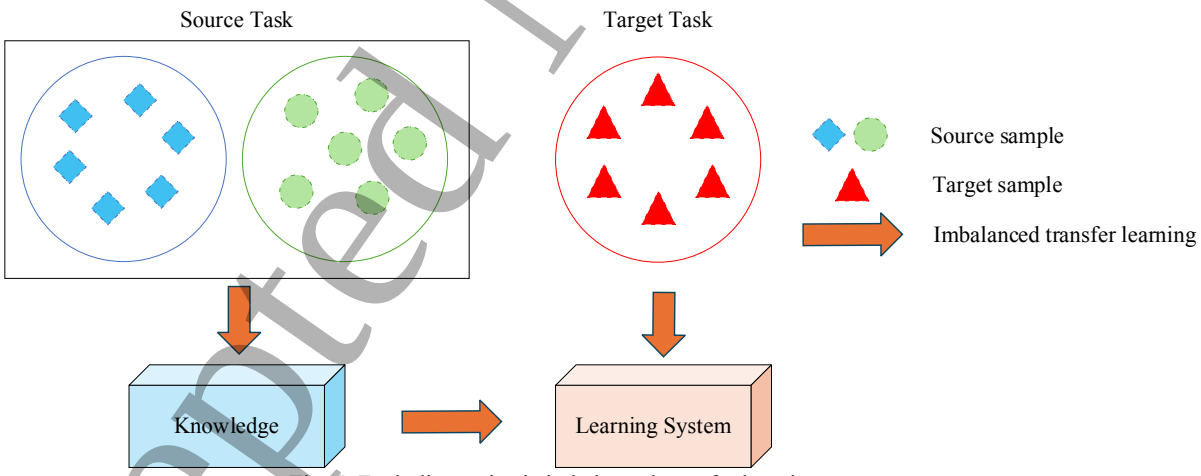


Fig. 5. Fault diagnosis via imbalanced transfer learning

3.2 Adversarial transfer networks

As shown in Fig. 6, the network consists of two symmetric structures: a source-domain feature extractor and a target-domain feature extractor, which minimize the model capacity and ensure the consistency of the feature mapping through the weight sharing mechanism; within the respective feature extractors, two feature fusion modules are introduced, which combine the depth-separable convolution with the global averaging and pooling to extract the local and global information in

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parallel, thus realizing a lightweight multiscale adversarial transfer network.

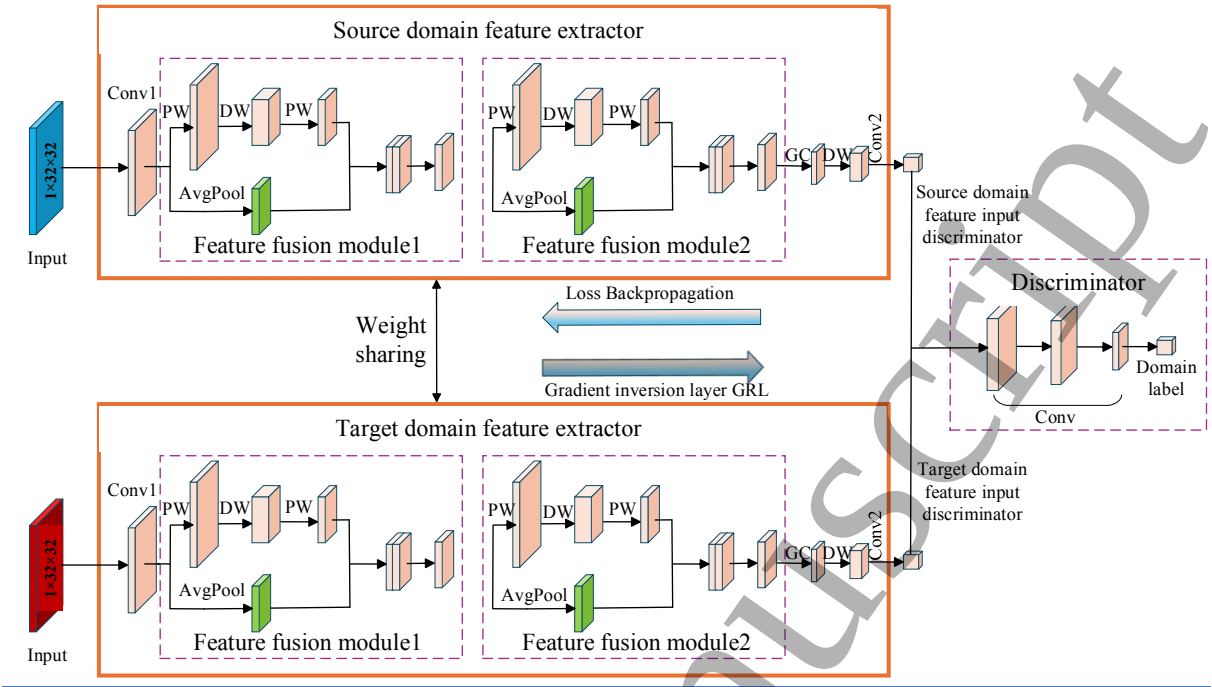


Fig. 6. Proposed methodological network framework

Inside the respective feature extractors, two feature fusion modules are introduced, combining deep separable convolution and global averaging, respectively, to extract local and global information in parallel, thus realizing lightweight multi-scale feature characterization. Meanwhile, with the help of the adversarial learning strategy of the gradient inversion layer and the domain discriminator, the extractors are forced to learn the domain-invariant features to eliminate the difference between the data distributions of the source domain and the target domain, which ultimately not only ensure the robustness of the cross-condition fault diagnosis but also take into consideration of the model's high efficiency of the inference ability.

3.2.1 Depthwise separable convolution

In depthwise convolution, each 3×3 convolution kernel is specifically applied to a single input channel, ensuring that each channel is convolved exclusively by its corresponding kernel. This results in the number of channels in the output feature map remaining identical to the input. The strict one-to-one kernel-channel mapping preserves the channel dimensionality, while the operation itself is analogous to standard convolution.

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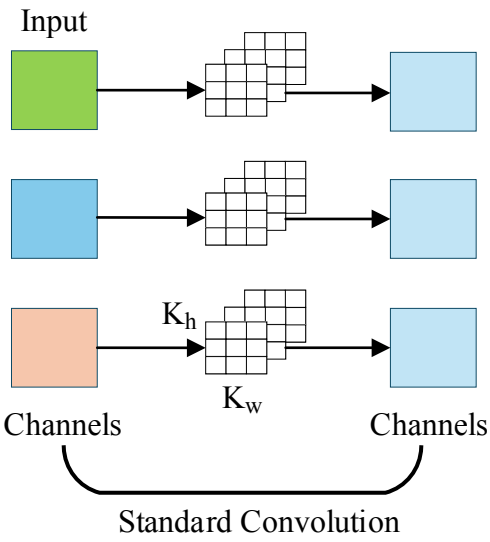


Fig. 7. Standard convolution network

Pointwise convolution follows, mixing channel-wise features to enable dimensionality adjustment and restore cross-channel information flow. As shown in Fig. 8, in depthwise convolution, each convolution kernel is dedicated to a single input channel, ensuring that each channel is convolved exclusively by its corresponding kernel.

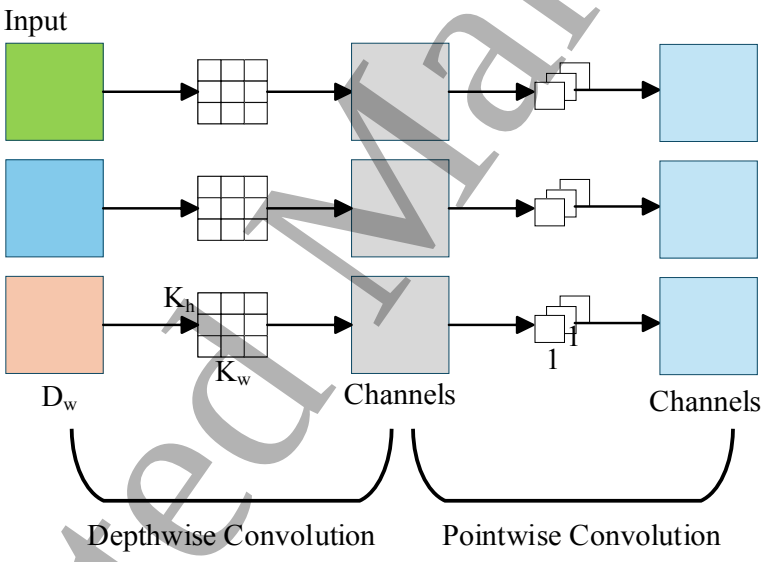


Fig. 8. Depthwise separable convolution network

The primary objective of pointwise is to address the insufficient feature combination capability of depthwise and the constraint on changing the number of channels. Assuming the input data is of size $D_h \times D_w$ with M channels and the standard convolution filter is of size $K_h \times K_w$ with N filters, Eq. (25) and Eq. (26) compute the standard convolution and depthwise separable convolution respectively

$$\zeta_d = D_h \times D_w \times K_h \times K_w \times M \times N \tag{25}$$

$$\zeta_s = D_h \times D_w \times K_h \times K_w \times M + D_h \times D_w \times M \times N \tag{26}$$

The computational ratio ζ_r between DSC and standard convolution is given by

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$$\zeta_r = \frac{\zeta_s}{\zeta_d} = \frac{1}{N} + \frac{1}{K_h \times K_w} \tag{27}$$

Depthwise separable convolution offers a significant reduction in computational cost, requiring approximately one-third the computational resources of standard convolution. This makes it an efficient alternative for reducing the overall computational load of a model.

3.2.2 Domain adversarial training

In the field of domain adaptation, adversarial domain adaptation stands out as an effective technique inspired by adversarial learning in generative adversarial networks (GANs) [34]. Throughout training, an adversarial interplay occurs between the feature extraction module tasked with producing domain invariant representations and the domain discriminator, which endeavors to differentiate source from target samples. This interplay continues until the distributions are significantly aligned, ideally reaching an approximate Nash equilibrium where the source-trained classifier can effectively generalize to the target domain. To achieve this, the model maximizes the domain classification, forcing the feature extractor to produce irrelevant representations. The loss function is formulated as follows

$$Loss = \mathrm{L}_s(\chi_S, Y_S) + \lambda \mathrm{L}_t(\chi_S, \chi_T) \tag{28}$$

Here, L_s represents the cross entropy loss computed on the source domain dataset, and L_t is the adversarial loss computed between the source and target domain datasets.

3.2.3 Feature fusion module

A lightweight feature fusion block was designed due to the computationally intensive and poor generalisation performance of traditional network models. The model is combined with an adversarial adaptation domain approach.The feature fusion block is shown in Fig. 9. The architecture of the proposed feature fusion module is meticulously designed to consist of two parallel branches. The first branch incorporates an average pooling layer, which effectively reduces spatial dimensions while retaining crucial information, thereby serving as a salient feature extractor. The second branch includes a depthwise separable convolution layer, which is a computationally efficient substitute for standard convolutional layers.

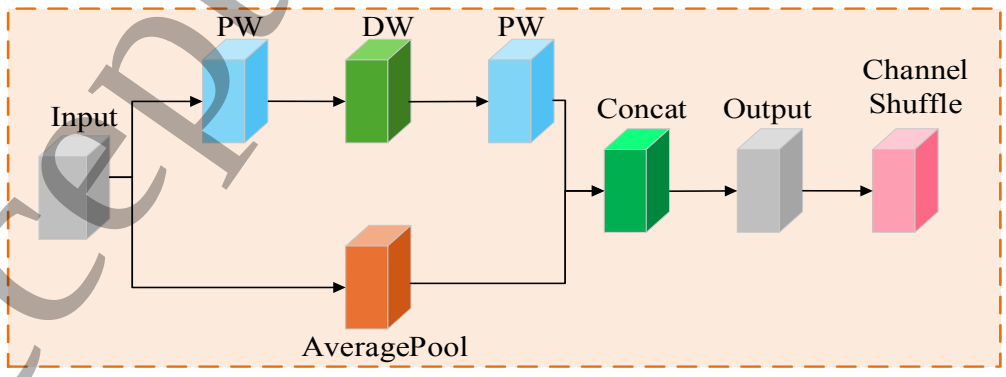


Fig. 9. Feature fusion module

The essence of the feature fusion mechanism hinges on the cohesive amalgamation of feature

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maps produced by both the average pooling layer and the depthwise separable convolution layer. Typically, augmenting the number of output channels necessitates an increase in the quantity of convolution kernel groups. Within this module, the ReLU6 activation function serves as the nonlinear activation unit [35]. ReLU6 not only effectively mitigates the risk of gradient saturation but also facilitates superior quantization and numerical operations on mobile devices with constrained resources. This particular activation function is instrumental in ensuring robust nonlinearity during feature extraction while simultaneously boosting the model's overall performance and its suitability for deployment in mobile environments.

$$\mathrm{Re}LU6 = \min(6, \max(0, x)) \tag{29}$$

ResNet [36] employs a residual learning technique to effectively reuse the high-level features, which enhances the feature extraction capability and improves classification accuracy. Similarly, this study effectively reuses high-level feature representations by incorporating a branch employing global average pooling and applying residual suppression mechanisms. When the feature fusion module is fused, channel crosstalk may hinder the interaction of feature information between two branches.

3.2.4 Diagnostic process and target optimization

Cross-domain fault diagnosis firstly requires the acquisition of vibration signals under different operating conditions, and then the source and target domains are divided, where the source domain dataset and the unlabelled target domain dataset are defined as $\Delta s = \{x_s^i, y_s^i\}(i = 1,2,3,\cdots n)$, $\Delta t = \{x_t^i, y_t^i\}(i = 1,2,3,\cdots n)$. For model training, the 1D data was shaped into channel 1 and 3D data of size 32 $\times$ 32 with two-channel inputs. Given that the classification features are directly introduced into the neural network $G_p(\cdot, \theta_p)$, it is unnecessary to decouple the feature extraction and classification processes. Consequently, the unified architecture that encompasses both the feature extraction and classification stages is symbolized as $G_{\mathrm{f}}, d(\bullet; \theta_f, d)$ and henceforth, this integrated module is designated as the feature extractor.

In the backpropagation phase, the parameters of the feature extractor and the domain classifier should be updated in distinct processes, with the label prediction loss L_y and domain classification loss L_d defined as

$$L_y = \mathsf{L}_s(G_f, d(x_i; \theta_f, d), y_i) \tag{30}$$

$$L_d = \mathsf{L}_{bc}(G_p(G_f, d(x_i; \theta_f, d), y_i), \theta_p, d_i) \tag{31}$$

where L_s is the cross-entropy loss, L_{bc} is the binary cross-entropy loss, y_i and d_i denote the true categorical and domain labels, respectively.

$$\vartheta = \frac{3 + \exp(-\varphi \bullet n)}{1 + \exp(-\tau \bullet n)} \tag{32}$$

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where, Eq. (32) ϑ is the equilibrium parameter of the loss function, n is the current number of iterations, $L_y^{\ i}$, $L_d^{\ i}$ denote the fault classification and domain classification losses of the $i-th$ sample, and the optimal values of the parameters can be expressed as

$$\arg\min \vartheta L_d(\arg\{\max \vartheta L_d(\theta_f, c)\}, \theta_p) \tag{33}$$

3.3 Overall framework for fault diagnosis

This is depicted in Fig. 10, the framework integrates dynamic correction based on digital twins and uncertainty modelling to develop a fault diagnosis method for cross-domain conditions and to identify defect scenarios. The primary steps are outlined below

(1) Data collection and pre-processing: Building high-fidelity models where sensors collect real-world data and DT models generate data to supplement missing labels.

(2) Cross-domain adaptation alignment：Cascade temporal and angular fourier transforms on the autocorrelation of an angle synchronized vibration signal to decouple speed variations and yield a sharp order frequency DCSC map of fault related periodic components.

(3) Data blending and enhancement: Fuses corrected analogue signals with real-world data to create a comprehensive dataset. The fusion process preserves the high fidelity of simulated data while embedding real-world variability and contingency factors, yielding a comprehensive, physically consistent dataset for robust training and validation of digital-twin models.

(4) The digital twin's dynamic calibration mechanism applies adaptive adjustment strategy for the simulated signals to minimize inter-domain shift and enforce coherence between simulated and empirical datasets according to distributional discrepancies between source and target domains, which results in the alignment between source and target domain data across diverse operating scenarios.

(5) Analysis of diagnostic results: The comprehensive training set is used as the input to DCLATN to explore the intrinsic correlation between vibration signals and their corresponding fault states; the generalisation performance of the network model is subsequently validated using the validation set at the end of each training round.

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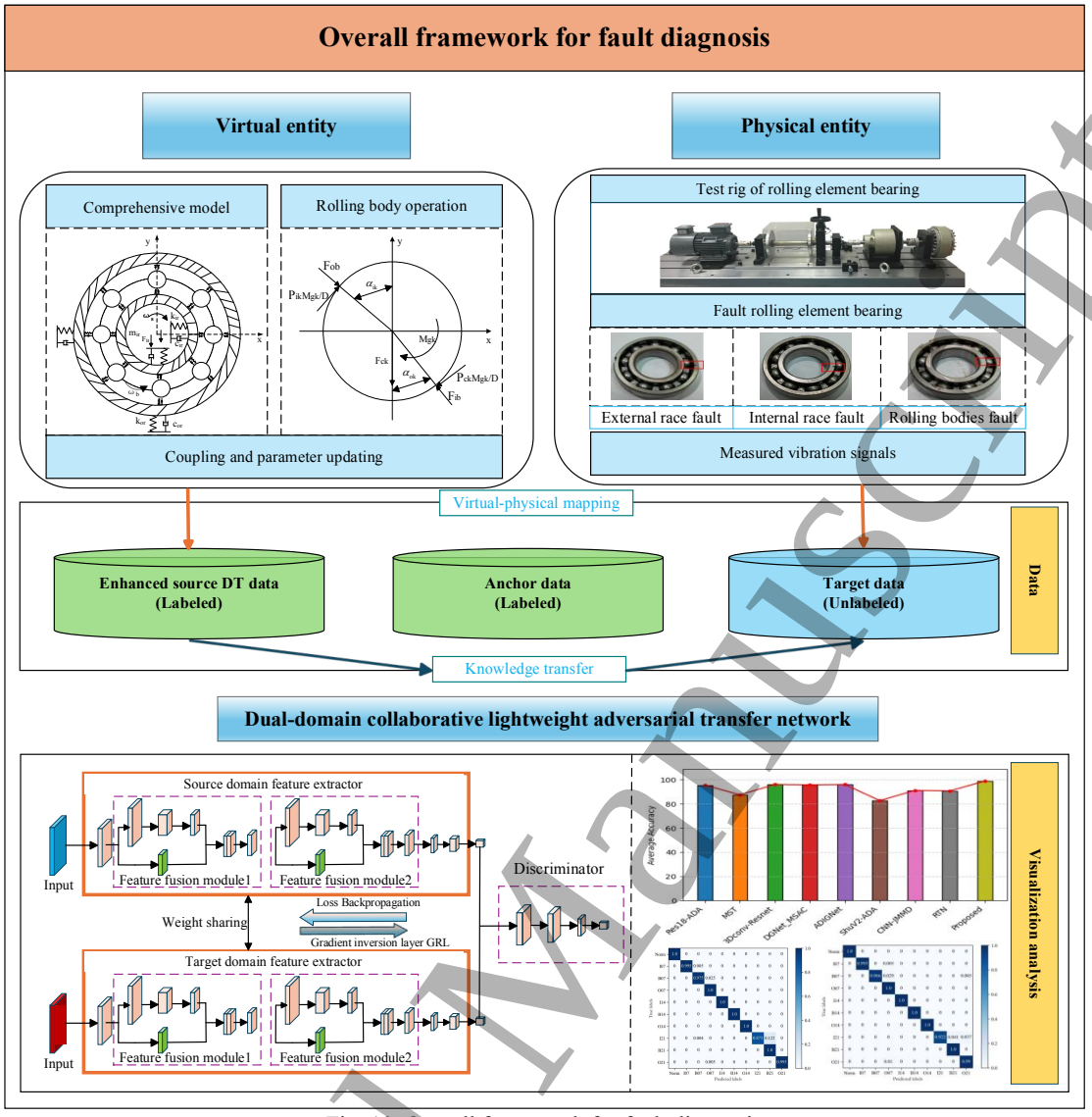


Fig. 10. Overall framework for fault diagnosis

4. Virtual model validation

A digital twin model of a rolling bearing is constructed in this paper for the purpose of conducting an in-depth study on its vibration characteristics. The model takes into account the friction effect as well as the influence of the rolling body stiffness on the simulated signals. Furthermore, the validity and accuracy of the digital twin model are verified by evaluating its time -domain and frequency-domain characteristics, reflecting the real world operating condition of rolling bearings. The digital twin model parameters listed in Table 1 can be used to complete model calibration and generate source domain datasets based on experimental data.

In this part, the accuracy of the virtual model in diagnosing rolling bearing faults under variable speed conditions is validated through a comparison of the simulated signals to actual vibration signals. The model simulates three conditions: defects in rolling bodies, internal raceway defects, and external raceway defects. Spring stiffness, damping ratios, and system mass were obtained through impulse-response characterization of a single-degree-of-freedom model. These values were then refined iteratively until the model's resonant frequency and amplitude replicated those

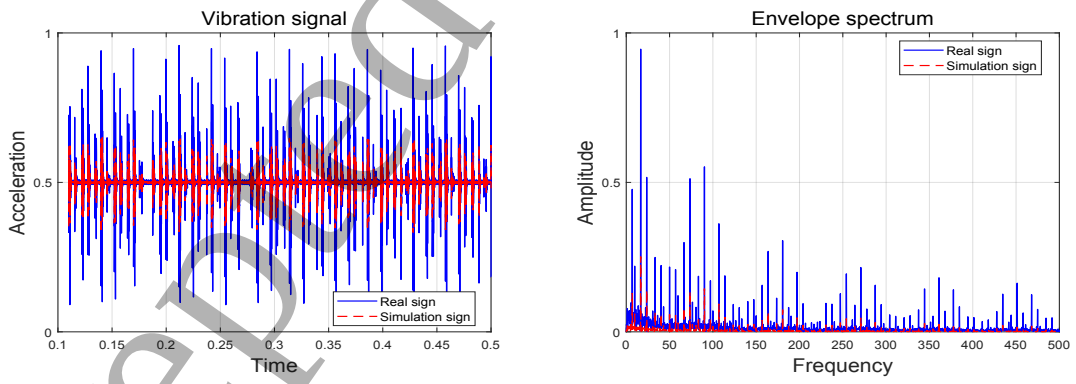
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observed experimentally.

Table1 Simulation parameters for SKF6205 bearing

SKF6205 Bearing Parameter	Value
external diameter	47mm
internal diameter	31mm
Width	15mm
Pitch diameter	39.04mm
Rolling bodies diameter	7.94mm
Number of balls	9
Basic static load ratings	14KN
Contact angle	0
Fault depth	0.1mm
The mass of the internal ring and the shaft	5kg
The mass of external ring and housing	10kg
Equivalent stiffness of internal ring and shaft	$7*10^7$N/m
Equivalent stiffness of external race and housing	$1.5*10^7$N/m
Radial force	20N
Damping coefficient of internal race	1500N·s/m
Damping coefficient of external race	2000N·s/m
Damping coefficient of rolling bodies	1500N·s/m

Fig. 11 shows the comparison between the simulated and real vibration signals for internal raceway faults, external raceway faults, and rolling-body faults for the Shandong University of Science and Technology dataset described in Section 5.1.1. For the internal ring defect, as shown in Fig. 10(a) and Fig. 10(b), the real measurements are highly consistent with the simulated signals, exhibiting the periodic pulsation characteristic of such faults. The theoretical internal raceway fault frequency calculated using Eq. (10) is 90.24 Hz, while the internal raceway fault frequency obtained from the envelope analysis of the twin data is 90 Hz. The error between the fault frequency of the twin data and the theoretical value is 0.26%, which indicates a high reliability.



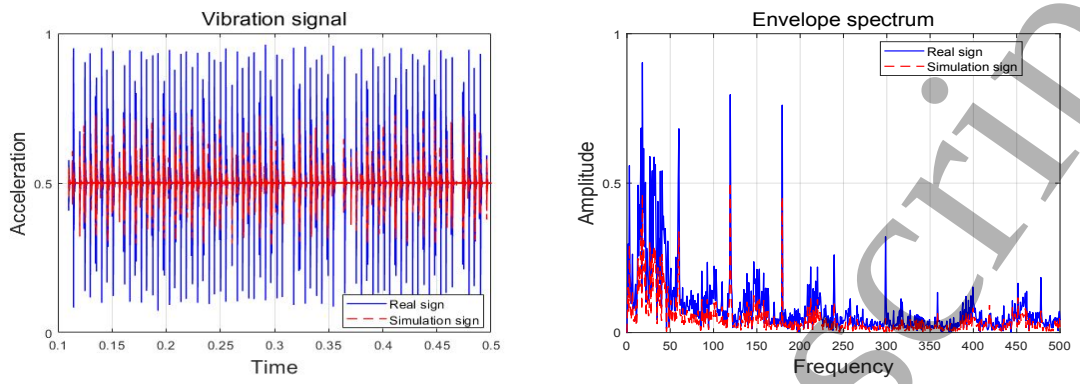
（a）Internal race defect vibration signal (b) Vibration signal envelope spectrum

Fig. 11. Internal ring defect data analysis

Fig. 12 illustrates the time-domain vibration waveform alongside its envelope spectrum for the twin-signal response of a rolling bearing containing an external raceway defect. When the external raceway of a rolling bearing fails, the defect location changes with the rotation of the external raceway. When the defect is located in the unloaded region, the shock peak is relatively small. Therefore, the time-domain waveform of the twin vibration signal exhibits alternating large and

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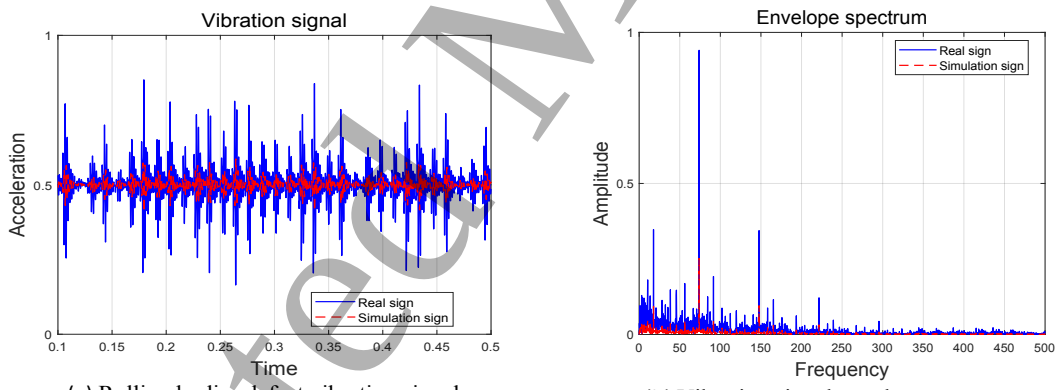
small shock peaks, which is consistent with the theoretical analysis. According to the external raceway fault characteristic frequency formula, the external raceway fault signature frequency is 59.75 Hz. The analog vibration signal shows a characteristic fault frequency of 60 Hz with an error of only 0.41% compared to the theoretical value.



(a) External race defect vibration signal (b) Vibration signal envelope spectrum

Fig. 12. Data analysis of external ring defects

The following Fig. 13 illustrates the time-domain waveforms and envelope spectra of the twin vibration signals associated with rolling bodies defects in a rolling bearing. For rolling bearings with rolling bodies defects, the rotation of the rolling bodies around the shaft center generates different levels of shocks at different locations. When the defective rolling bodies is in the loading zone, a significant shock peak is generated. When the rolling bodies moves out of the loading zone, the shock peak caused by the defect decreases. This leads to a vibration signal time-domain waveform characterized by alternating high and low shock peaks.



(a) Rolling bodies defect vibration signal (b) Vibration signal envelope spectrum

Fig. 13. Rolling body defect data analysis

By observing the above figure, it can be observed that under the conditions of internal raceway defects, external raceway defects, and rolling body defects, the simulated vibration signals exhibit vibration periods and shock pulse characteristics that are congruent with those of the actual vibration signals in the time domain, while the characteristic frequency of the fault also closely corresponds. It has been demonstrated that the proposed model can accurately replicate the bearing fault vibration signals.

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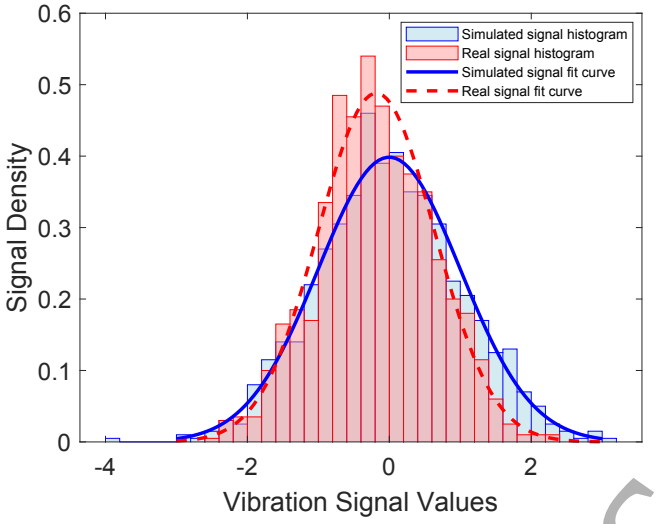


Fig.14. Real signal density versus simulated signal density for inner-ring faults

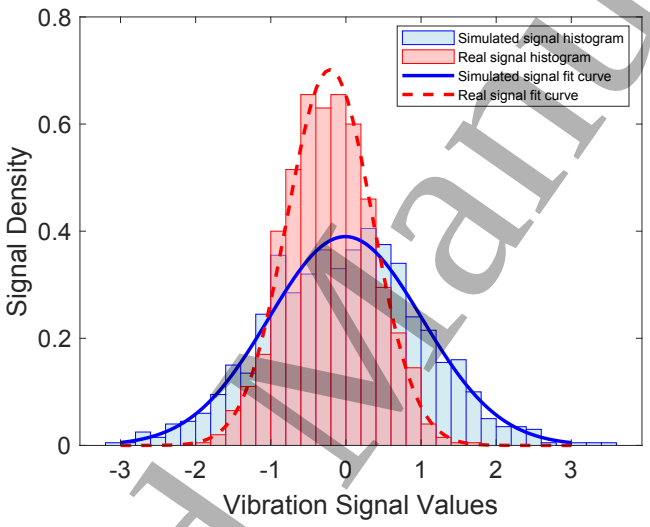


Fig.15. Real signal density versus simulated signal density for outer-ring faults

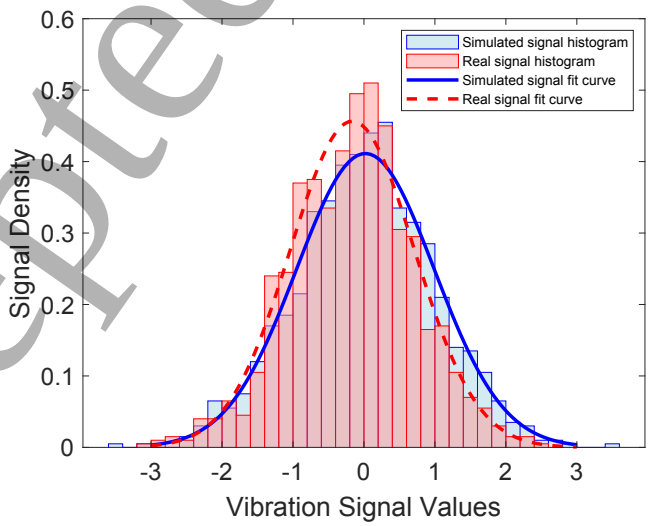


Fig.16. Real signal density versus simulated signal density for rolling bodies faults

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Table2 Comparison of real collected data with simulated data

Type		Effective value	Skew
Inner-ring faults	Collect data	0.725g	9.120g^2
	Simulated data	0.718g	8.980g^2
	Error	1.06%	1.5%
Outer-ring faults	Collect data	0.810g	9.760g^2
	Simulated data	0.802g	9.580g^2
	Error	1.02%	1.9%
Rolling bodies faults	Collect data	0.780g	9.450g^2
	Simulated data	0.768g	9.280g^2
	Error	1.5%	1.8%

A comparative analysis was performed on fault data acquired by physical sensors and fault data generated via simulation. Specifically, Figure 14 depicts the waveform comparison between the measured signal and the simulated inner ring fault signal, whereas Figure 15 presents the contrast between the collected actual signal and the simulated outer ring fault signal. In addition, Figure 16 illustrates the alignment between the real-world collected signal and the simulated rolling element fault signal. It can be observed that the two types of signals (measured and simulated) exhibit negligible differences in signal density, with their mean values and standard deviations also remaining in close agreement.

5. Experimental comparison

5.1 Case 1: Fault diagnosis based on the SDUST dataset

5.1.1 Bearing test bench introduction

The physical entity serves as the foundation for constructing digital twin models. The experimental equipment used in this study is the bearing fault diagnosis test bench from Shandong University of Science and Technology [37]. The main components of this test bench are shown in Fig. 17, including a 1.5 KW inverter motor, coupling shaft, rotor, supporting bearing, magnetic vibration sensor, and brake. Vibration signals from the bearing group are collected and recorded in the X and Y directions at a sampling frequency of 25.6 kHz by magnetic vibration sensors installed on the bearings. The test bearings are of type SKF6205 and feature four health conditions, as shown in Fig. 17 internal raceway fault, external raceway fault, and rolling bodies fault. Each fault category includes three different fault sizes: 0.2 mm, 0.4 mm, and 0.6 mm. Data is collected at four different rotational speeds: 1500 rpm, 2000 rpm, 2500 rpm, and 3000rpm.

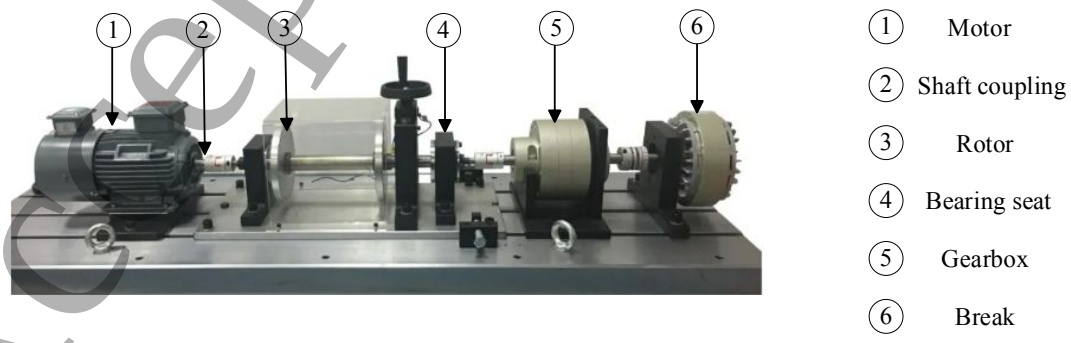
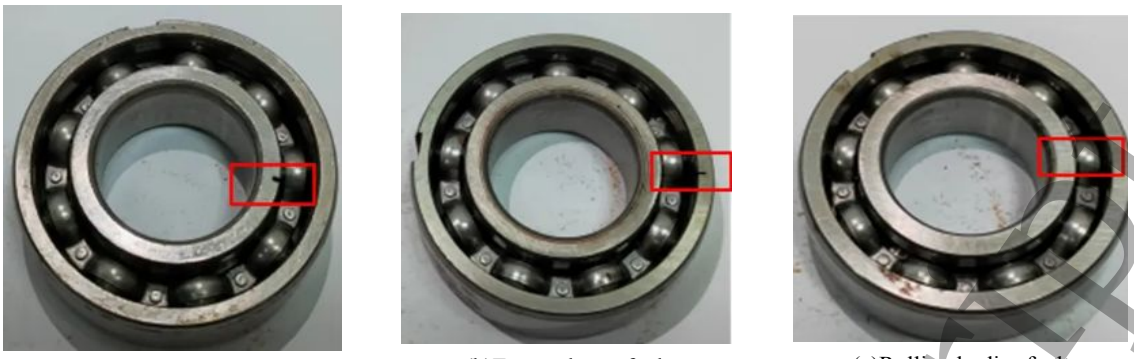


Fig. 17. Shandong University of Science and Technology bearing fault diagnosis test bench

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(a)Internal race fault (b)External race fault (c)Rolling bodies fault

Fig .18. Bearings with different fault types

To ensure comparability of experiments, each simulation is conducted with the bearing operates at a constant speed, torque, and radial force under different experimental parameters. As shown in the table below, the Shandong University of Science and Technology dataset comprises four operating conditions. The operational parameters include speed, radial force, and fault size. The test bench operates at four rotational speeds: n=1500 rpm, n=2000 rpm, n=2500 rpm, and 3000 rpm, with a radial force of F = 20 N. To simulate local defects in bearings, electro-engraving is used to introduce 0.4 mm-wide indentations in the internal raceway, external raceway, and rolling bodies, representing three types of faults.

Table3 Operating parameters of shandong university of science and technology bearing test platform

Condition	Speed (rpm)	Radial force (N)	Fault width (mm)
0	1500	20	0.4
1	2000	20	0.4
2	2500	20	0.4
3	3000	20	0.4

5.1.2 Comparison analysis and parameter settings

Experiments were performed using the bearing dataset from Shandong University of Science and Technology to validate the proposed method. The experiment was run on the Intel i5-10700K CPU and the NVIDIA RTX 4050 GPU. The neural network structure starts from the signal layer, with 1 output channel and an output size of 32×32. Then, after passing through the convolution layer, the output channel increases to 48, and the output size becomes 16×16. Enter feature extraction module 1, first pass through PW layer 1 and then through DW layer 1. The output size is 8×8, then there is PW layer 2 and avgpool layer 1. The output size is 4×4. Finally, the output channels are integrated into 96, and the output size is 8×8. In the feature extraction module 2, after passing through the PW layer 3 and the DW layer 2, PW layer 4 and Avgpool layer 2, the output channel integration is 192. The output size is 4×4. Subsequently, there are also the global convolution layer (GC), and the DW layer 3 . Output size 2×2 and PW layer 5.

For DCLATN, data fusion at the feature level is performed using a cascade model with the parameter settings shown in Table 4. Furthermore, for the DCLATN model, the learning rate is configured to 0.0004, and the batch size is set to be 64. To assess the efficacy of the DCLATN approach, eight state-of-the-art deep learning algorithms have been chosen as benchmark

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comparison methods, including Res18-ADA (ResNet18 Adversarial Domain Adaptation) [38], MST (Multi-scale transform) [39], 3Dconv-Resnet (3D-Dynamic convolution-based residual network) [40], DGNet_MSAC (Domain generalization network with multidomain specific auxiliary classifiers) [41], ADIGNet (Adversarial domain-invariant generalization Network) [42], ShuV2-ADA (ShuffleNetV2 Adversarial Domain Adaptation) [43], RTN (Residual Transfer Networks) [44] and CNN-JMMD (Convolution neural network with joint maximum mean discrepancy) [45].

Table 4 Network structure parameter Settings

Structure	Convolution kernel size	Stride	Padding	Output channel	Output size
Signal	-/-	-/-	-/-	1	32×32
Conv	3×3	2	1	48	16×16
PW	1×1	1	0	48	16×16
DW	3×3	2	1	48	8×8
PW	1×1	1	0	48	8×8
Avgpool	3×3	2	1	48	8×8
Feature extraction Module 1 channel	96	8×8			
PW	1×1	1	0	96	8×8
DW	3×3	2	1	96	4×4
PW	1×1	1	0	96	4×4
Avgpool	3×3	2	1	96	4×4
Feature extraction Module 2 channel	192	4×4			
GC	3×3	1	1	96	4×4
DW	3×3	2	1	96	2×2
PW	1	2	0	10	1×1

In the experimental part of this study, the effect of Batch size on the diagnostic results of DCLATN was explored in depth using Task C as the object of study. To minimize the potential interference of random factors on the diagnostic results of the experiment, this experiment was conducted with 10 independent iterations for each batch size setting. Ultimately, the experimental results were measured based on the accuracy of all repeated trials. In the initial experiment, the batch size was set to 64 while fixing the learning rate to 0.0004 for the fault detection task using DCLATN. The results of the experiment are shown in Fig. 19. After examining a wide range of batch sizes, it is found that the diagnostic accuracy is maintained above 90% when the batch size is varied within a certain interval. However, when the batch size was further increased to 256, a more significant decrease in diagnostic accuracy was observed. Subsequent findings show that the DCLATN model can achieve the highest diagnostic accuracy when the batch size is 64, a finding that provides an important reference basis for optimization of model parameters.

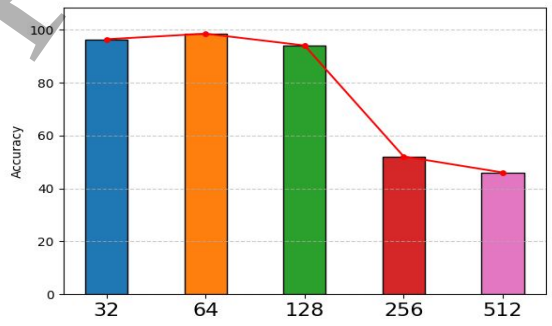


Fig. 19. Bitch size

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5.1.3 Comparative results with other methods

In this section of the experiment, the parameter settings of the baseline method and the parameter choices of the proposed method align with those of Experimental Group A, so they are not repeated here. As shown in table 5,in the fault diagnosis test conducted under two distinct working conditions, 90% of the samples are consistently allocated to the training set, while the remainder are designated for the test set. The evaluation metric employed is the mean fault diagnosis accuracy derived from 10 experimental iterations. The diagnostic results under the variable speed condition are shown in Figs. 20-23 while more detailed results are recorded in the table.

Table 5 Experimental configuration for transfer learning tasks

Transfer task name	Source domain dataset	Target domain dataset	Test dataset
Task A	1500 r/min	2000 r/min	2000 r/min
Task B	1500 r/min	2500 r/min	2500 r/min
Task C	1500 r/min	3000 r/min	3000 r/min

The diagnostic results reveal that the average fault diagnosis accuracy of the proposed method reaches 98.93% under the variable speed conditions of Tasks A, B and C. It is particularly noteworthy that most of the methods show a significant decrease in the fault diagnosis performance at Task C compared to Tasks A and B. The results of the proposed methods are summarized in Table 6 below. This declining pattern likely arises from the increased heterogeneity and complexity of the vibration data acquired under the sequential deceleration acceleration protocol (Task C). Such recordings are more prone to exhibiting shifting characteristics than those obtained solely during acceleration, thereby compounding the difficulty of accurate diagnosis.

As shown in Table 6, In this experiment, the average fault diagnosis accuracy of the RTN model is 90.80%, which is better than the traditional residual network data fusion model but still 8.13% lower than the proposed method. The average diagnosis accuracy of the MST model is 87.60%. The average diagnosis accuracy of the ShuV2-ADA model is 82.77%. The average diagnosis accuracy of the DGNet_MSAC model is 95.86%.In addition, the differences between the diagnostic performances of Res18-ADA, and ADIGNet are relatively small, where 3Dconv-Resnet achieves the highest diagnostic accuracy of 96.06%. Nevertheless, a 2.83 % performance shortfall persists relative to the DCLATN algorithm. In Task C, Res18-ADA attains a mean classification accuracy of 93.5%, outperforming all other benchmarked techniques. CNN-JMMD has the lowest standard deviation, indicating strong robustness, but the latter exhibits superior robustness compared to DCLATN. To further explore the merits of the proposed method, feature visualization experiments were conducted. Specifically, the t-distributed stochastic neighbor embedding (t-SNE) algorithm was adopted to visualize the deep features extracted by the feature extractor module. Figure 20 depicts the t-SNE visualization results of distinct methods on the test dataset. It can be observed that 3Dconv-Resnet, MST, RTN, and the proposed method all suffer from feature confusion to varying extents across different health states. Nevertheless, the proposed method achieves notably superior classification performance compared with the other

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three approaches.

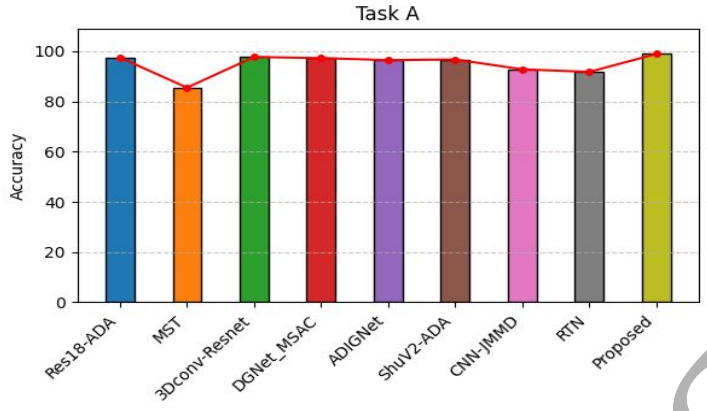


Fig. 20. Assessing fault diagnosis accuracy using various methods under Task A

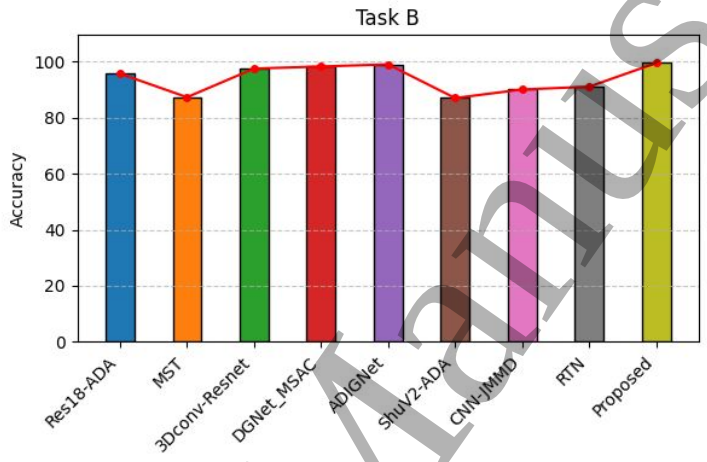


Fig. 21. Assessing fault diagnosis accuracy using various methods under Task B

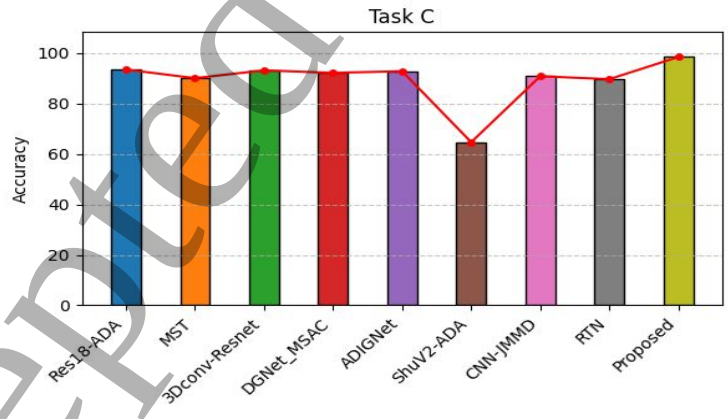


Fig. 22. Assessing fault diagnosis accuracy using various methods under Task C

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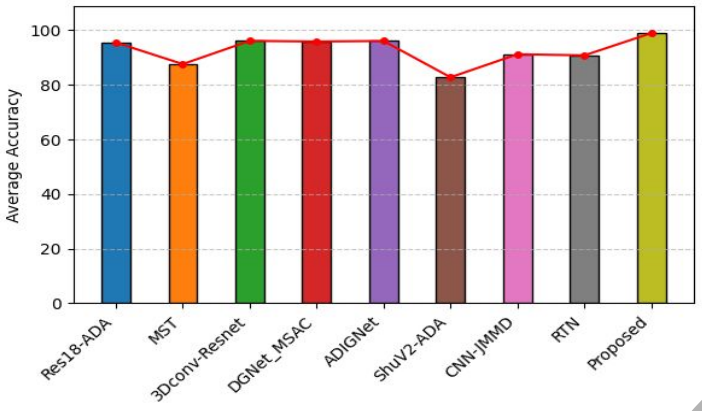
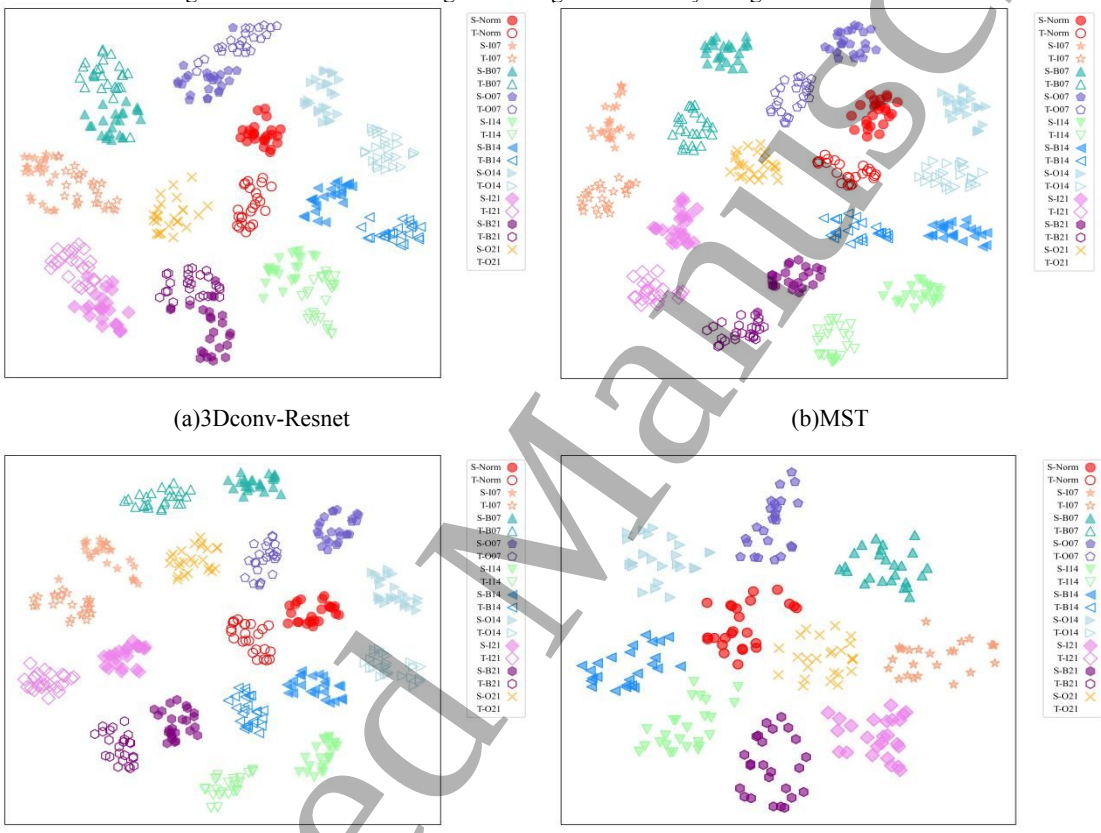


Fig. 23. Assessment of average fault diagnosis accuracy using different methods



(a)3Dconv-Resnet (b)MST

(c)RTN (d)Proposed

Fig.24. Feature visualization of different models

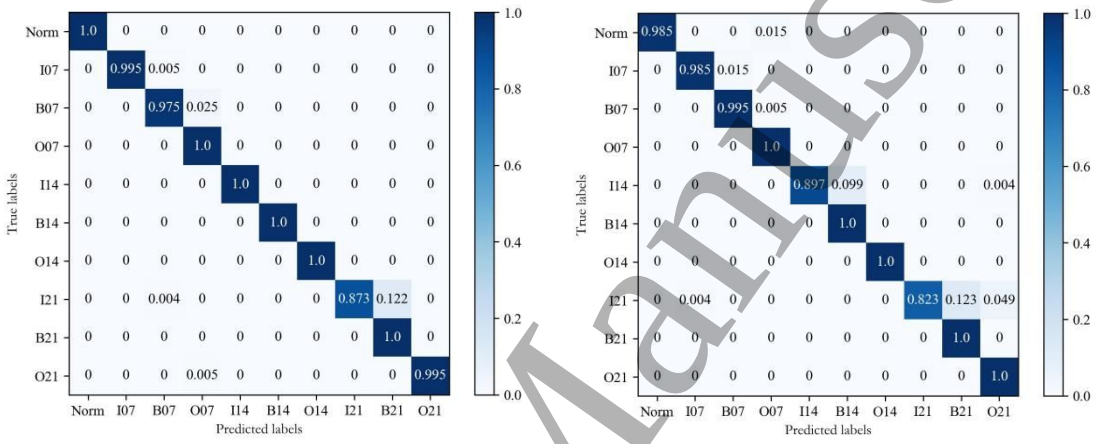
Table 6 Diagnosis results from the dataset of Shandong University of Science and Technology

Model	Task A	Task B	Task C	Average accuracy rate
Res18-ADA	97.45 %	95.60 %	93.50 %	95.51 %
MST	85.50 %	87.20 %	90.10 %	87.60 %
3Dconv-Resnet	97.70 %	97.40 %	93.20 %	96.10 %
DGNet_MSAC	97.25 %	98.15 %	92.20 %	95.86 %
ADIGNet	96.45 %	98.90 %	92.85 %	96.06 %
ShuV2-ADA	96.70 %	86.90 %	64.70 %	82.77 %
CNN-JMMD	92.80 %	89.95 %	90.85 %	91.20 %
RTN	91.75 %	90.95 %	89.70 %	90.80 %
Proposed	98.90 %	99.40 %	98.50 %	98.93 %

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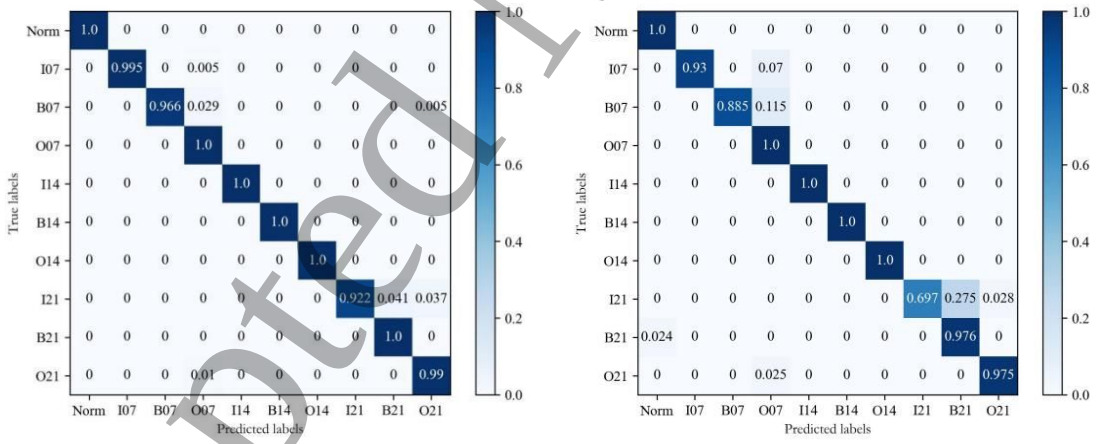
5.1.4 Confusion matrix analysis

By examining the confusion matrix in Fig. 25, we can conclude that in Task B, the DCLATN algorithm only misclassified two samples with actual category 7 as category 8, while the 3Dconv-Resnet algorithm showed multiple instances of misclassification of samples, which suggests that compared to our proposed DCLATN algorithm, the 3Dconv-Resnet has obvious shortcomings in classification performance. Further, by examining the confusion matrix in Fig. 26, in Task C, the 3Dconv-Resnet algorithm has significantly lower classification accuracy than the DCLATN algorithm in the identification of fault categories 2 and 7. Based on the above analysis results, we can clearly conclude that the DCLATN algorithm significantly outperforms 3Dconv-Resnet in terms of classification performance.



(a)Confusion matrix obtained by the DCLATN (b)Confusion matrix obtained by the 3Dconv-Resnet

Fig. 25. Confusion matrix of Task B



(a)Confusion matrix obtained by the DCLATN (b)Confusion matrix obtained by the 3Dconv-Resnet

Fig. 26. Confusion matrix of Task C

5.2 Case 2: Fault diagnosis based on the SCU dataset

5.2.1 Experiment platform introduction

The bearing fault diagnosis dataset was developed in the laboratory of Soochow University [46]. The experiment utilized an SKF6205-2RS model bearing, which consists of 9 rolling bodies. Each rolling bodies has a diameter of 7.9400 mm, and the pitch circle diameter is 39.0398 mm. The experimental setup is illustrated in Fig. 27.

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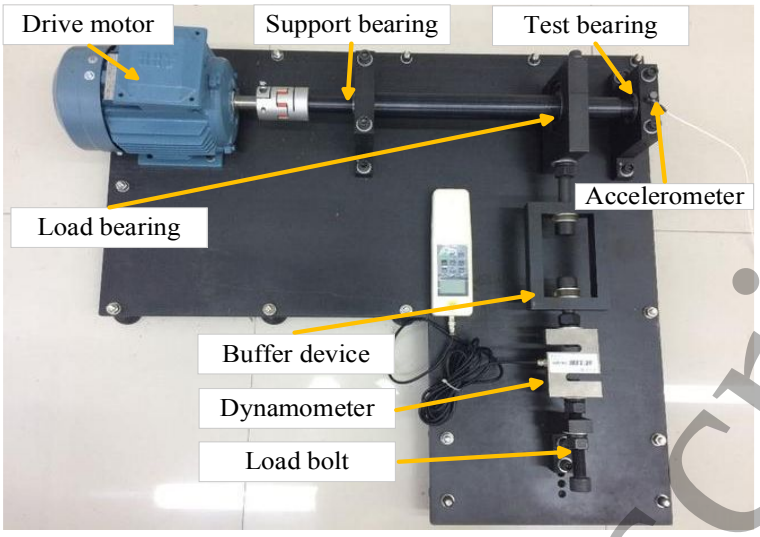


Fig. 27. SCU experiment platform

The experimental platform is made up of a prime mover motor, a plummer coupling, a healthy bearing, a test bearing, an accelerometer, and other necessary devices. To investigate cross-domain fault diagnosis, we introduced three types of faults on the test bearing's surface: internal raceway, ball, and external raceway. Fault sizes were set to 0.2 mm, 0.4 mm, and 0.6 mm respectively, and the created using the wire cutting method. Data were collected under four load conditions: 0 kN, 1 kN, 2 kN, and 3 kN. The specific parameters are outlined in Tables 8 and 9.

Table 7 SCU datasets settings.

Form	Spinning speed/ (rpm)	Torque load / (kN)	Number of sample
1	961	0	2000
2	961	1	2000
3	961	2	2000
4	961	3	2000

Table 8 Transfer task settings.

Task name	Source domain dataset	Target domain dataset	Test dataset
Task a	0 kN	1 kN	1 kN
Task b	0 kN	2 kN	2 kN
Task c	0 kN	3 kN	3 kN

5.2.2 Model comparison and analysis

In this experiment, given the significant increase in diagnostic difficulty of the SCU dataset compared to the SDUST dataset, the experimental training was configured to run for 3000 iterations, while the other hyper-parameter configurations were kept the same as in the first case. Table 9 presents the detail the performance of the different models in the cross-domain diagnosis task. It is worth noting that the DCLATN model demonstrates significantly better diagnostic performance than other methods, despite the fact that it employs a very small number of parameters and computational resources, which thoroughly proves the exceptional capability of the DCLATN model in complex diagnostic scenarios.

Table 9 illustrates the diagnostic results obtained for variable speed conditions. More detailed results are recorded in the table. For Tasks a, b, and c, diagnostic assessments demonstrate that the proposed approach delivers a mean fault-diagnosis accuracy of 98.70%. This declining trend may

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be due to the increased difference in data distribution between Tasks a and b. These signals are more likely to exhibit offsets, which increase the diagnostic difficulty. In this experiment, the RTN model reaches an average accuracy of 89.43% in diagnosing faults, which is better than the traditional residual network data fusion model, but still 9.27% lower than the proposed method. The MST model has an average accuracy of 77.61% in diagnosis. In addition, the diagnostic performance differences among Res18-ADA , 3Dconv-Resnet and ADIGNet are relatively small , among which Res18-ADA has the highest diagnostic accuracy of 96.21% , but still has a gap of 2.49% compared with the proposed method DCLATN. Specifically, the average accuracy provided by CNN-JMMD is 93.5%, which is the highest among all the compared methods in Task c. 3Dconv-Resnet has the smallest standard deviation, which indicates that it is robust, but the latter exhibits stronger robustness compared to DCLATN.

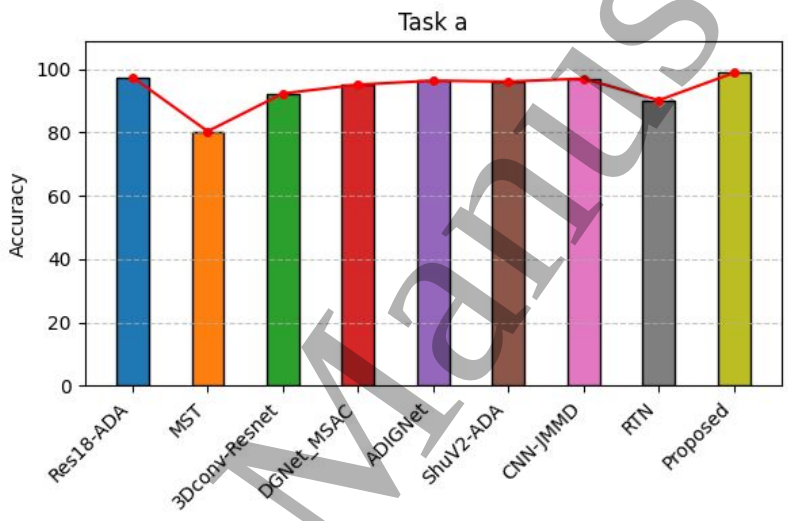


Fig. 28. Assessing fault diagnosis accuracy using various methods under Task a

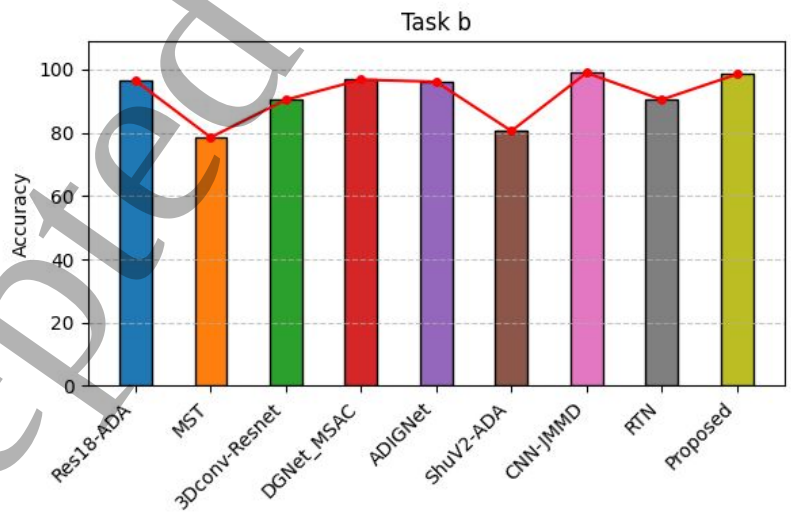


Fig. 29. Assessing fault diagnosis accuracy using various methods under Task b

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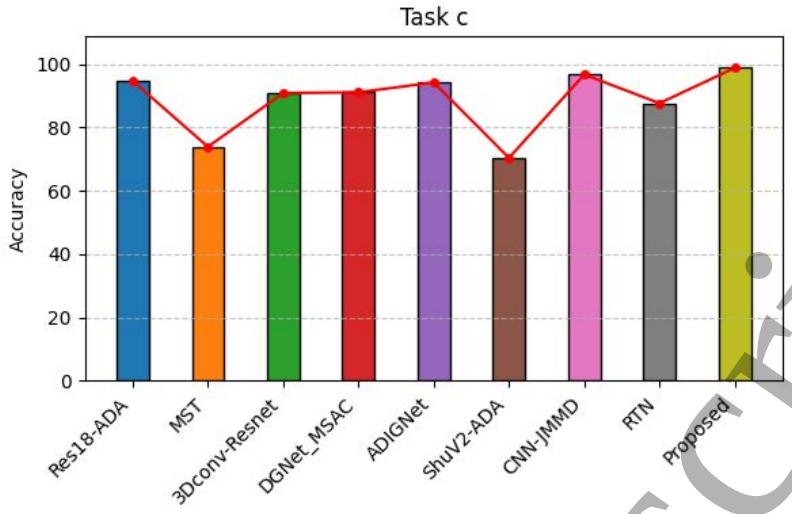


Fig. 30. Assessing fault diagnosis accuracy using various methods under Task c

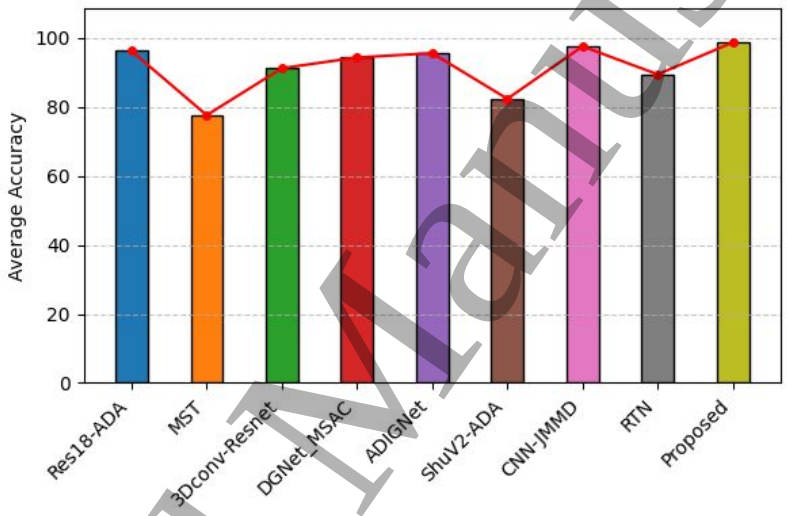
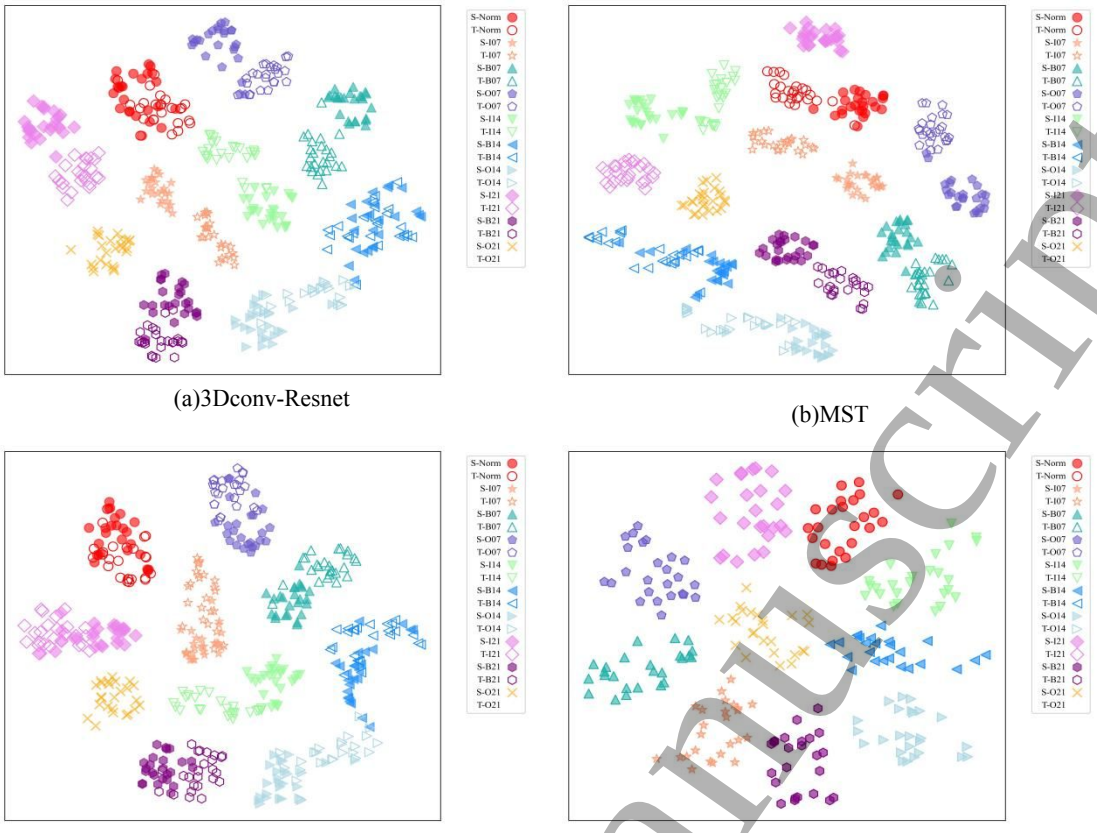


Fig. 31. Assessment of average fault diagnosis accuracy using different methods

Table 9 Diagnosis results from the dataset of SCU

Model	Task A	Task B	Task C	Average accuracy rate
Res18-ADA	97.50 %	96.45 %	94.70 %	96.21 %
MST	80.40 %	78.55 %	73.90 %	77.61 %
3Dconv-Resnet	92.35 %	90.40 %	90.85 %	91.20 %
DGNet_MSAC	95.15 %	96.75 %	91.10 %	94.33 %
ADIGNet	96.40 %	96.10 %	94.20 %	95.57 %
ShuV2-ADA	96.10 %	80.55 %	70.40 %	82.35 %
CNN-JMMD	97.05 %	98.90 %	96.70 %	97.55 %
RTN	90.20 %	90.45 %	87.65 %	89.43 %
Proposed	98.85 %	98.45 %	98.80 %	98.70 %

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(a)3Dconv-Resnet (b)MST

(c)RTN (d)Proposed

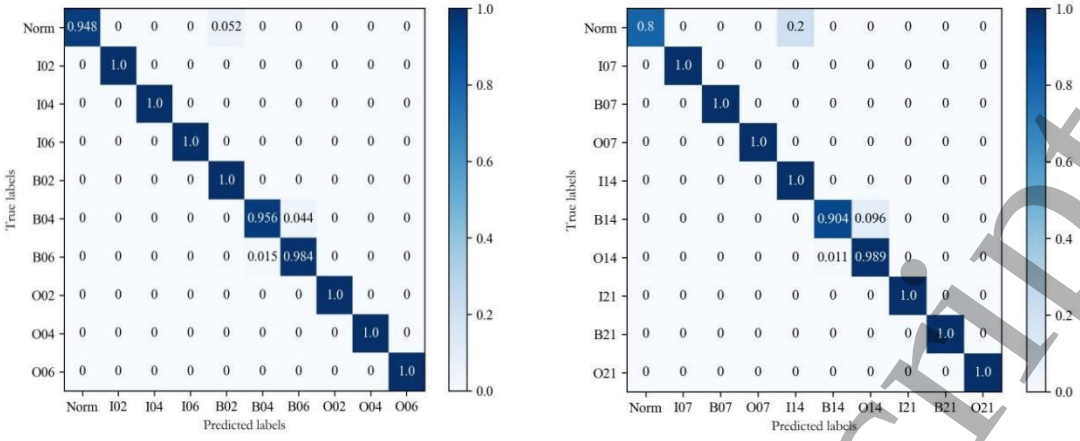
Fig.32. Feature visualization of different models

5.2.3 Confusion matrix analysis

To thoroughly investigate the cause of this phenomenon, this paper provides a deeper analysis of the confusion matrices of the three migrated tasks, displayed in Figs. 33-35. Specifically, the confusion matrices associated with Tasks a, b, and c are shown in Figs. 33-35 (a), (b), and (c), respectively. In Task c, the ADIGNet model exhibits significant confusion in distinguishing between normal state (Nor), B04 and B06 fault types. This confusion worsens as the Task b progresses. Fig. 34 (b) clearly shows that although ADIGNet is able to effectively recognize ball bearing faults, it has significant deficiencies in accurately classifying smaller fault types (e.g. B04 and B06). In addition, ADIGNet also frequently misdiagnoses normal states (NOR) as ball bearing faults. This phenomenon may stem from the increased discrepancy in data distribution between Tasks a and c.

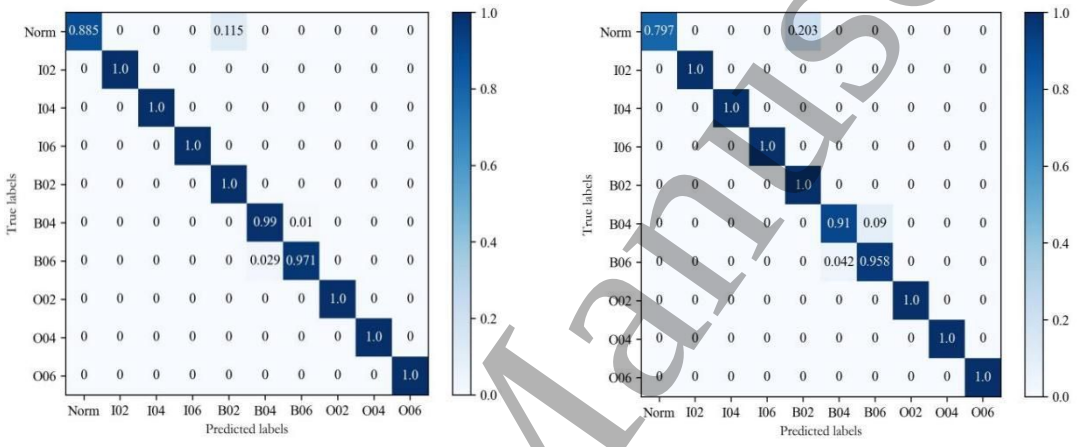
Moreover, evaluation indicates that the feature fusion enabled DCLATN algorithm upholds superior diagnostic precision even in intricate transfer scenarios featuring a limited number of fault classes. This result indicates that the DCLATN algorithm has significant advantages in extracting domain-invariant features, which effectively improves the robustness and accuracy of the model in cross-domain diagnosis tasks.

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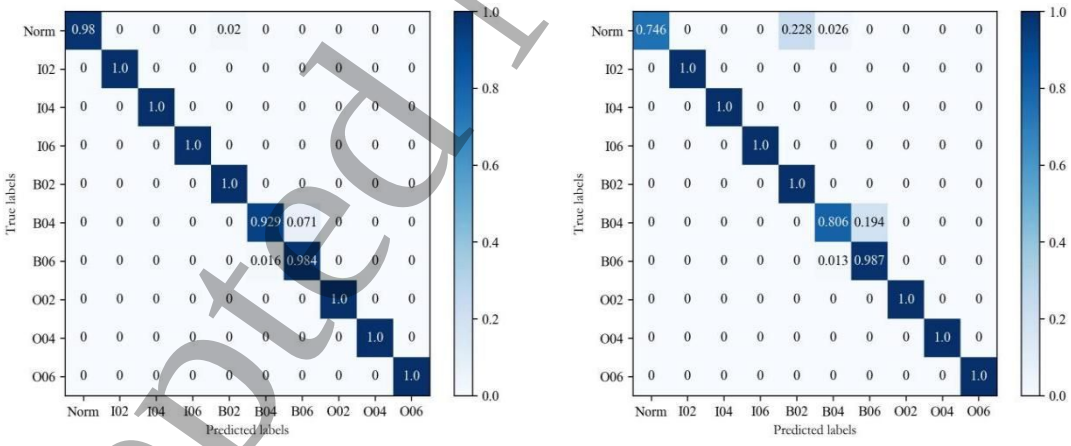
(a)Confusion matrix obtained by the DCLATN (b)Confusion matrix obtained by the ADIGNet

Fig.33. Confusion matrix of Task a



(a)Confusion matrix obtained by the DCLATN (b)Confusion matrix obtained by the ADIGNet

Fig. 34. Confusion matrix of Task b



(a)Confusion matrix obtained by the DCLATN (b)Confusion matrix obtained by the ADIGNet

Fig. 35. Confusion matrix of Task c

5.2.4 Ablation experiment

The ablation experiments are mainly aimed at verifying two key conclusions. First, the feature-fusion strategy markedly enhances the model's capacity for feature extraction, as illustrated by ablating the average-pooling operation within the fusion module. Second, the integration of feature maps produced via average pooling substantially improves the model's adaptability, as demonstrated by substituting that pooling operation with a 1×1 convolutional

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layer.

The first ablation model is ADAD-DC, and the second ablation model is ADAMP, and the results of the various ablation experiments, as well as the model's lightweight metrics, are detailed in Table 10. Due to the lack of a feature fusion strategy, the diagnostic performance of ADAD-DC starts to deteriorate significantly when performing task B, and the deterioration is especially severe when performing task C. The performance of ADAD-DC is not as good as that of ADAD-DC, which is not as good as that of ADAD-DC. However, if the channel cascading strategy is abandoned and a convolutional layer is used instead to extend the number of channels, ADAMP is still able to achieve feature fusion and achieves more satisfactory diagnostic results when performing the transmission task. However, the model size of ADAMP is relatively large, which is mainly due to the use of convolutional layers in its left branch. Experimental results show that DCLATN not only significantly reduces the number of model parameters and computation time, but also significantly improves the generalization ability of the model.

To comprehensively evaluate the effectiveness and computational complexity of the proposed method, this study compares its computational efficiency with baseline models across three key dimensions—floating-point operations per second (FLOPs), number of parameters, and inference time using the SDUST and SCU datasets. Table 10 presents the computational complexity comparison between the ADAD-DC, ADAMP modules, and DCLATN. The ADAD-DC model achieves FLOPS values and parameter counts of $0.215*10^6$ million and 0.19 million respectively, representing approximately 1.5 times that of the baseline model. DCLATN achieved FLOPS and parameter counts of $0.341*10^6$ M and 0.26M respectively, representing an approximately 27% improvement over the baseline ADAD-DC model. These results demonstrate that the proposed method achieves significant accuracy gains without substantially increasing computational complexity.

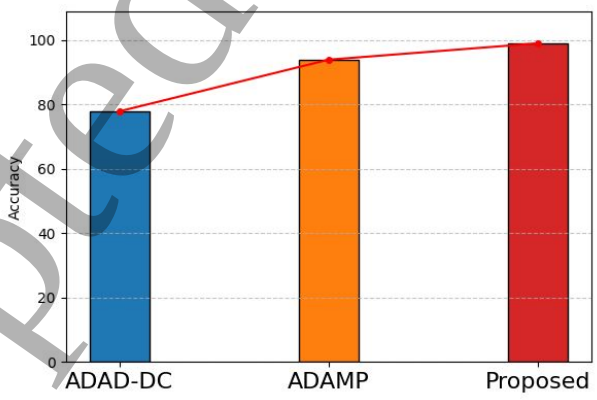


Fig.36. Ablation experiment data comparison

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Table 10 Experimental results and model lightweighting metrics

Model	Task A	Task B	Task C	Average	Params (MB)	Flops ($\times 10^6$)
ADAD-DC	97.80 %	80.10 %	55.60 %	77.83 %	0.26	0.341
ADAMP	98.20 %	97.60 %	85.75 %	93.85 %	0.23	0.249
DCLATN	98.90 %	99.40 %	98.50 %	98.93 %	0.19	0.215

6. Conclusion

In conclusion, this study addresses the challenge of low diagnostic accuracy in rolling bearing fault diagnosis caused by distribution discrepancies between virtual and physical data due to external factors such as load and rotational speed fluctuations. To tackle this issue, an intelligent diagnostic framework integrating an adaptively updated high-fidelity digital twin model and a DCLATN is proposed, aiming to enhance cross-domain adaptability under complex operating conditions. The core contributions include developing an advanced digital twin architecture that combines physical modeling with data-driven methods, enabling the transfer of simulation insights to real-world scenarios with time-varying dynamics, alleviating the scarcity of annotated datasets, and improving model robustness across operational conditions; formulating a high-fidelity multi-physics vibration model based on digital twin technology to generate synthetic datasets covering diverse fault modes under nonlinear regimes, applicable to various bearing designs and industrial contexts; and proposing the DCLATN, which integrates gradient inversion layers and discriminators to reduce data distribution discrepancies, enhance fault feature invariance, and enable efficient cross-domain feature extraction. Experimental results validate that the proposed framework outperforms state-of-the-art methods, ensuring robust and accurate cross-domain fault diagnosis by eliminating domain disparities in vibration signals, with significantly higher accuracy compared to traditional approaches. This research provides a promising paradigm for integrating digital twins and transfer learning in mechanical fault diagnosis, with future work focusing on optimizing performance under extreme conditions and expanding applications to other rotating machinery to enhance industrial practicality.

Declaration of conflicts of interest

The author(s) declared no potential conflicts of interest withrespect to the research, authorship, and/or publication of thisarticle.

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